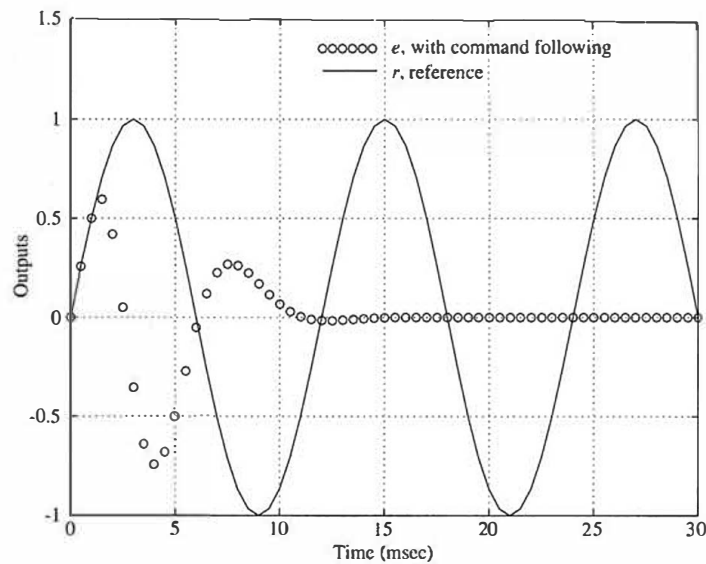


Figure 8.33
Response of disk drive to
sinusoidal reference
following as in Fig. 8.32,
Example 8.20



marked by small circles in Fig. 8.33. Note that, once $\bar{p}$ has converged, there is no noticeable following error.

8.6 Effect of Delays

Many feedback control systems have a pure time delay, λ , imbedded in some part of the loop. A one cycle delay was analyzed in Section 7.3.2 by adding a z^{-1} to the system model and using the root locus method to show that the stability of the system was decreased when no change to the compensation was made. We can also analyze the effect of delays with frequency response methods as in Section 7.4 by reducing the phase by $\omega\lambda$. This analysis easily shows that, if no changes are made to the compensation, the phase margin and hence stability will be reduced. Either design method can be used to modify the compensation so the response is improved.

For state-space design, we saw in Section 4.3.4 that, for an actuator delay, one state element must be added to the model for each cycle of delay or fraction thereof. In other words, a delay of $0 < \lambda \leq T$ will require an increase in the order of the system model by 1, a delay of $T < \lambda \leq 2T$ will increase the order by 2, and so on. Using the pole placement approach, we can assign any desired pole locations to the system. Therefore, we are able to achieve the same closed-loop poles in a system with delays as one without delays; however, there are extra

poles in the system with delays. Those extra poles can slow down the response even if they are selected as fast as possible, that is, at $z = 0$. The response of the system to command inputs and disturbances is also affected by the location of the delays, that is, whether they are part of the control actuator or the sensor. Figure 8.34 shows systems with the delays in the two locations.

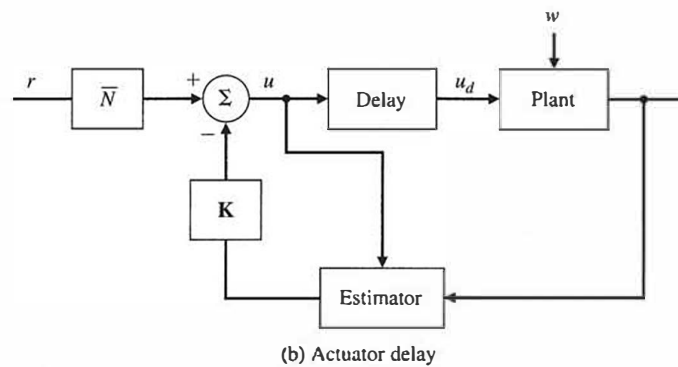
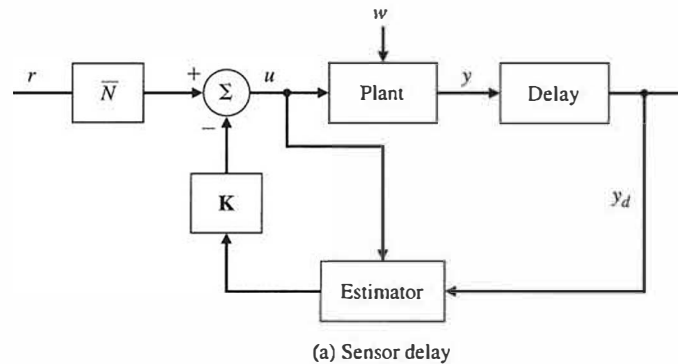
8.6.1 Sensor Delays

For the sensor delay, Fig. 8.34(a), an estimator can be used to reconstruct the entire state; therefore, the undelayed state is available for control. The system can be made to respond to command inputs, r , in exactly the same way that a system would respond without a sensor delay because the estimator sees the command through the feedforward of u and does not depend on the delayed output to detect it. Therefore, no estimator error is excited and the undelayed state estimate is accurate. On the other hand, a disturbance input, w , will not usually be seen by the estimator until the output, y_d , responds; therefore, the estimator error will be increased by the delay.

delay model

Figure 8.34

System with delays,
(a) sensor delay,
(b) actuator delay



The model of a one cycle delay of a quantity y is

$$y_{1d}(k+1) = y(k), \quad (8.92)$$

where y_{1d} is the delayed version of y and is an additional state element that is added to the system model. The model for more than one cycle can be obtained by adding more similar equations and state elements. So, for two cycles of delay, we would add

$$y_{2d}(k+1) = y_{1d}(k), \quad (8.93)$$

to Eq. (8.92), where y_{2d} is one more state element and is the value of y that is delayed by two cycles. Therefore, for a system given by

$$\begin{aligned} \mathbf{x}(k+1) &= \mathbf{\Phi}\mathbf{x}(k) + \mathbf{\Gamma}u(k) \\ y(k) &= \mathbf{H}\mathbf{x}(k), \end{aligned}$$

the system model including a two-cycle sensor delay is

$$\begin{bmatrix} \mathbf{x}(k+1) \\ y_{1d}(k+1) \\ y_{2d}(k+1) \end{bmatrix} = \begin{bmatrix} \mathbf{\Phi} & \mathbf{0} & \mathbf{0} \\ \mathbf{H} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{1} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ y_{1d}(k) \\ y_{2d}(k) \end{bmatrix} + \begin{bmatrix} \mathbf{\Gamma} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix} u(k), \quad (8.94)$$

$$y_d(k) = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{1} \end{bmatrix} \begin{bmatrix} \mathbf{x}(k) \\ y_{1d}(k) \\ y_{2d}(k) \end{bmatrix}, \quad (8.95)$$

where y_d is the output y delayed by two cycles. Any number of cycles of delay can be achieved easily by using this scheme.

If a sensor had delay that was not an integer number of cycles, it would not influence the sampled value until the next sample instance. Therefore, sensor delays must be an integer number of samples.

Note that, due to the column of zeros, the augmented system matrix, $\mathbf{\Phi}_a$ in Eq. (8.94) will always be singular, a fact that will cause difficulties when calculating gains for a current estimator using `acker` in MATLAB.¹⁷

◆ Example 8.21 *Effect of a Sensor Delay on the Satellite Attitude Response*

Examine the step response to a command input r of the satellite attitude control with one cycle of delay at the sensor. Place two of the control poles at $z = 0.8 \pm 0.25j$, as was the case for Examples 8.2 and 8.12, and the additional pole for the delay state at $z = 0$. Place the poles for a prediction estimator at $z = 0.4 \pm 0.4j$, as was the case for Examples 8.5 and 8.12, and the additional pole for the delay state at $z = 0$. Compare the results with Example 8.12, the same system step response without a delay.

¹⁷ We will see in Chapter 9 that a singular $\mathbf{\Phi}$ matrix will also cause difficulty with `dlqr`.

Solution. The state-space equations for this system are (from Eqs. 8.94 and 8.95)

$$\begin{bmatrix} x_1(k+1) \\ x_2(k+1) \\ y_{1d}(k+1) \end{bmatrix} = \begin{bmatrix} 1 & T & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ y_{1d}(k) \end{bmatrix} + \begin{bmatrix} T^2/2 \\ T \\ 0 \end{bmatrix} u(k),$$

$$y_d(k) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(k) \\ x_2(k) \\ y_{1d}(k) \end{bmatrix}.$$

Use of `acker` in MATLAB with the augmented Φ and Γ matrices as defined above with the desired control pole locations yields

$$K = [10.25 \quad 3.4875 \quad 0].$$

The first two elements are the exact same values that were obtained in Examples 8.2 and 8.12, and the zero third element means that the delayed state element is ignored. Thus the sensor delay has no effect on the controller.

Use of `acker` in MATLAB with the augmented Φ and H matrices as defined above with the desired estimator pole locations yields

$$L_p = \begin{bmatrix} 1.72 \\ 5.2 \\ 1.2 \end{bmatrix}.$$

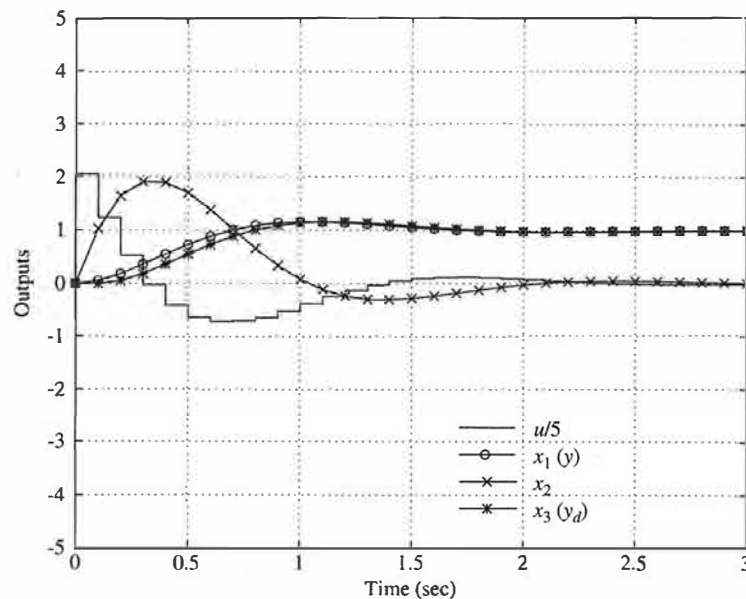
In evaluating the step response according to Eq. (8.79), $\bar{N}$ needs to be evaluated with the new values of N_x and N_u . In this case, Eq. (8.73) shows that

$$N_x = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad N_u = 0, \quad \text{and} \quad \bar{N} = 10.25.$$

The use of step with the system defined by Eq. (8.79) produces the step response shown in Fig. 8.35. It includes the output y and the delayed output y_d . Note that the response of y is precisely the same as the response for Example 8.12 in Fig. 8.17 even though the only quantity fed back was the delayed output. This result follows because the current value of the state was estimated correctly due to the feedforward from the command input and the current value of the state was used by the controller.

It is remarkable that a system with a sensor delay can respond to a command input in precisely the same way that the undelayed system would respond. This implies that the system closed-loop transfer function was such that the pole at $z = 0$ was not excited. The ability to design such a control system is enhanced by the state-space controller/estimator approach. It is important to note, however, that if the system had a disturbance, w , that was not fed into the estimator in the feedforward path, the delayed output measurement would cause a delayed estimator response and, thus, a degraded response of the system to that disturbance.

Figure 8.35
Step response for input
command for Example
8.21



application to engine
control

An example of a system in common use with a sensor delay is the fuel injection control for an automobile engine. Here, the sensed value of the fuel-air ratio in the exhaust is delayed by the piston motion and the time for the exhaust stream to reach the sensor. The primary disturbance is the motion of the throttle by the driver's foot; however, this motion can be sensed and used as feedforward to the estimator. Thus the estimator structure is capable of instantaneous rejection of the throttle disturbance in spite of the significant delay of the exhaust sensor. (See Fekete, 1995.)

8.6.2 Actuator Delays

The model for an actuator delay is derived in Section 4.3.4. The result is Eq. (4.79), which allows for any length delay, fractional or whole cycles. The poles of a system with actuator delays can be placed arbitrarily, just like the sensor delay case, and it makes sense to add a pole at $z = 0$ for each added delay state element. However, for input commands or disturbance inputs, the extra poles at $z = 0$ will be excited and the system response will be delayed by the actuator delay. This fact can be understood by examining Fig. 8.34(b). There is no way that the input r can influence the plant without passing through the delay. A disturbance w will influence the plant without delay; however, the feedback through the estimator will be delayed before it can counteract the disturbance, thus causing an increased sensitivity to disturbances.

◆ **Example 8.22** *Effect of an Actuator Delay on an Engine Speed Governor*

An engine speed governor consists of an rpm sensor, usually a device that counts the time between the passage of teeth on a gear, and logic to determine the input to the fuel injector (diesel engines) or to the throttle actuator or spark advance (gasoline engines). Figure 8.36 shows the block diagram including the engine model, which is a simple first order lag with a time constant of around 3 sec so that $a = 0.3$. Fig. 8.36(a) shows the system without any delay, the ideal case. The actuator delay shown in Fig. 8.36(b) varies from engine to engine. For a gasoline engine with a throttle actuator, it is typically around a half engine cycle due to the piston motion (100 msec at 600 rpm). For diesel engines the actuator can be made to act more quickly; however, some manufacturers use injectors that have a full engine cycle delay (200 msec at 600 rpm). For this example, we will use a sample time of $T = 200$ msec and a one cycle actuator delay.

Investigate the response of the system for step input commands, r , and impulsive disturbances, w . Compare the results for control systems structured as in the three cases in Fig. 8.36. Pick the control gain so that the poles are at $z = 0.4$ for the ideal case and the estimator gain (if there is one) so that its pole is at $z = 0.2$. Select $\bar{N}$ so that there is no steady-state error.

Solution. The step responses are shown in Fig. 8.37. For the ideal case (Fig. 8.36(a)), the gains were found to be

$$K = 9.303 \quad \text{and} \quad \bar{N} = 10.303.$$

The step response for $r = 1$ is shown in Fig. 8.37(a) and was obtained using step.

The model of the system with the delay in Fig. 8.36(b) can be accomplished with the aid of Eq. (4.79). For this system it reduces to

$$\begin{bmatrix} x(k+1) \\ u_d(k+1) \end{bmatrix} = \begin{bmatrix} \Phi & \Gamma \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x(k) \\ u_d(k) \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k) + \begin{bmatrix} \Gamma \\ 0 \end{bmatrix} w(k),$$

$$y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x(k) \\ u_d(k) \end{bmatrix},$$

where Φ and Γ can be found using MATLAB's `c2d`, but can also be computed easily for this first order case to be

$$\Phi = e^{-aT} \quad \text{and} \quad \Gamma = 1 - e^{-aT}.$$

Figure 8.37(b) shows the response of this system with the classical feedback as shown in Fig. 8.36(b) using the same gains as in (a). Not only is the response delayed by a cycle, but it has become oscillatory. A z -plane analysis of this system explains why: the roots are at $z = 0.47 \pm 0.57j$, which yield an equivalent s -plane damping of $\zeta = 0.33$.

Figure 8.37(c) shows the response of the system with the estimator structure as shown in Fig. 8.36(c) using the desired control pole locations of $z = 0.4, 0$. This yielded gains of

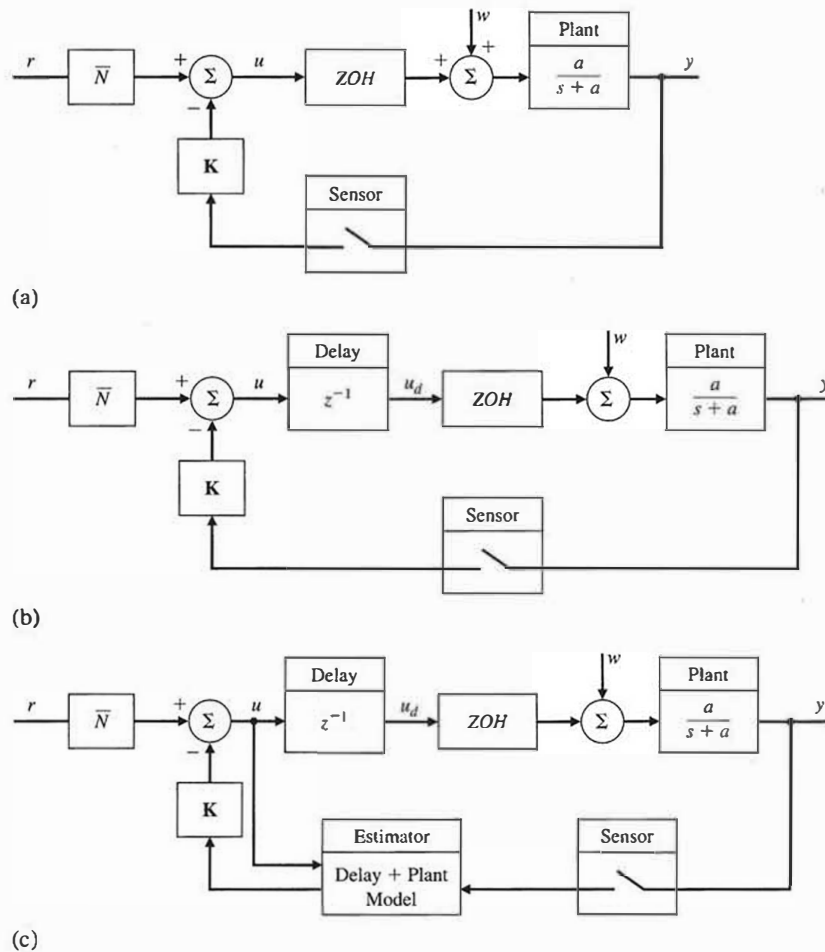
$$K = [8.76 \quad 0.54] \quad \text{and} \quad \bar{N} = 10.303.$$

The estimator poles were placed at $z = 0.2, 0$, which yielded an estimator gain of

$$L_p = \begin{bmatrix} 0.742 \\ 0 \end{bmatrix}.$$

Figure 8.36

Engine speed control block diagrams, (a) the ideal feedback system without a delay and without an estimator, (b) the classical feedback with an actuator delay, (c) the estimator structure with an actuator delay



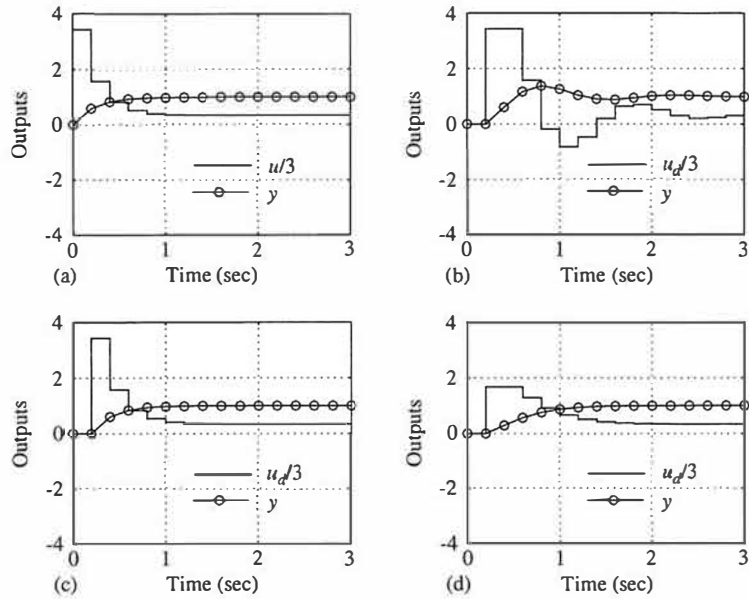
The response was obtained using Eq. (8.79) and is identical to the ideal case; however, it is delayed by one cycle. As previously discussed, this delay cannot be reduced if its source is due to the actuator.

The oscillatory response shown in Fig. 8.37(b) also can be eliminated by reducing the gain in the classical structure of Fig. 8.36(b), an easier solution than adding an estimator. Reducing the gain, K , from 9.3 to 4.0 produced the response in Fig. 8.37(d). While the oscillations have been removed, the response has slowed down. There is no value of gain in this simple structure that will duplicate the quality of the response obtained from the estimator system, although the difference is not large for this example.

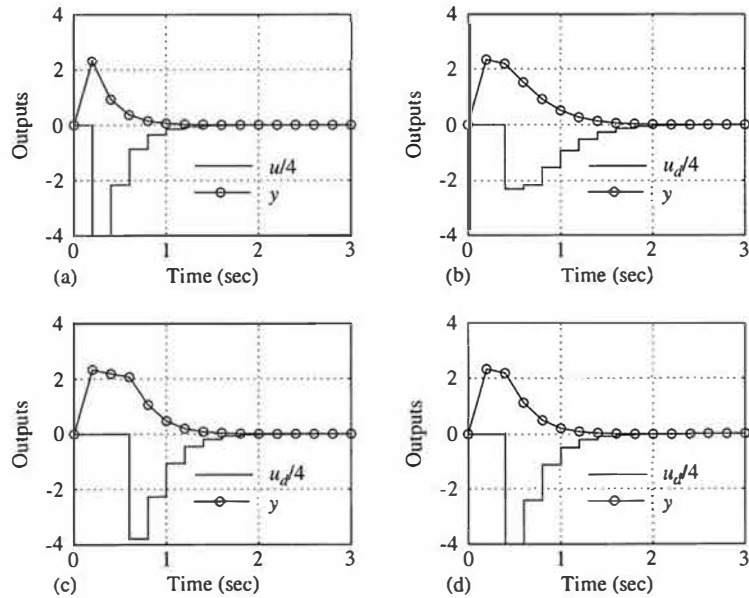
Three of the cases were analyzed for an impulsive disturbance, w and shown in Fig. 8.38(a, b, and c). Qualitatively, the same kind of responses were obtained; however, the predictor estimator had an extra cycle of delay compared to the classical feedback system; therefore, there is little difference between the responses in Figs. 8.38(b) and 8.38(c). A current estimator

Figure 8.37

Responses to a step input, r , for Example 8.22. (a) Ideal case, no delay ($K = 9.3$), (b) classical feedback with delay ($K = 9.3$), (c) predictor estimator with delay, (d) classical feedback with delay ($K = 4.0$)

**Figure 8.38**

Responses to impulsive disturbance, w , for Example 8.22. (a) Ideal case, no delay ($K = 9.3$), (b) classical feedback with delay ($K = 4$), (c) predictor estimator w/delay ($K = 9.3$), (d) current estimator w/delay ($K = 9.3$)



formulation eliminates the extra delay and improves the response, which was obtained by evaluating Eq. (8.81) and is shown in Fig. 8.38(d).

Example 8.22 shows that an actuator delay will effect the response to a command input or a disturbance regardless of the control implementation. This is an inherent characteristic of an actuator delay for any system. However, use of an estimator including the delay minimizes its effect. Use of a current estimator provides the best response to a disturbance.

8.7 *Controllability and Observability

Controllability and observability are properties that describe structural features of a dynamic system.¹⁸ These concepts were explicitly identified and studied by Kalman (1960) and Kalman, Ho, and Narendra (1961). We will discuss only a few of the known results for linear, constant systems that have one input and one output.

We have encountered these concepts already in connection with design of control laws and estimator gains. We suggested in Section 8.1.3 that if the matrix C given by

$$C = [\Gamma : \Phi\Gamma : \dots : \Phi^{n-1}\Gamma]$$

is nonsingular, then by a transformation of the state we can convert the given description into the control canonical form and construct a control law such that the closed-loop characteristic equation can be given arbitrary (real) coefficients. We begin our discussion of controllability by making the definition (the first of three):

controllability definitions

I. The system (Φ, Γ) is controllable if for every n th-order polynomial $\alpha_c(z)$, there exists a control law $u = -Kx$ such that the characteristic polynomial of $\Phi - \Gamma K$ is $\alpha_c(z)$.

And, from the results of Section 8.1.3, we have the test:

The pair (Φ, Γ) is controllable if and only if the rank of $C = [\Gamma : \Phi\Gamma : \dots : \Phi^{n-1}\Gamma]$ is n .

The idea of pole placement that is used above to define controllability is essentially a z -transform concept. A time-domain definition is the following:

¹⁸ This section contains material that may be omitted without loss of continuity.

II. The system (Φ, Γ) is controllable if for every $\mathbf{x}_0$ and $\mathbf{x}_1$ there is a finite N and a sequence of controls $u(0), u(1), \dots, u(N)$ such that if the system has state $\mathbf{x}_0$ at $k = 0$, it is forced to state $\mathbf{x}_1$ at $k = N$.

In this definition we are considering the direct action of the control u on the state $\mathbf{x}$ and are not concerned explicitly with modes or characteristic equations. Let us develop a test for controllability for definition **II**. The system equations are

$$\mathbf{x}(k+1) = \Phi \mathbf{x}(k) + \Gamma u(k),$$

and, solving for a few steps, we find that if $\mathbf{x}(0) = \mathbf{x}_0$, then

$$\mathbf{x}(1) = \Phi \mathbf{x}_0 + \Gamma u(0),$$

$$\mathbf{x}(2) = \Phi \mathbf{x}(1) + \Gamma u(1)$$

$$= \Phi^2 \mathbf{x}_0 + \Phi \Gamma u(0) + \Gamma u(1),$$

$$\vdots$$

$$\mathbf{x}(N) = \Phi^N \mathbf{x}_0 + \sum_{j=0}^{N-1} \Phi^{N-1-j} \Gamma u(j)$$

$$= \Phi^N \mathbf{x}_0 + [\Gamma : \Phi \Gamma : \dots : \Phi^{N-1} \Gamma] \begin{bmatrix} u(N-1) \\ \vdots \\ u(0) \end{bmatrix}.$$

If $\mathbf{x}(N)$ is to equal $\mathbf{x}_1$, then we must be able to solve the equations

$$[\Gamma : \Phi \Gamma : \dots : \Phi^{N-1} \Gamma] \begin{bmatrix} u(N-1) \\ u(N-2) \\ \vdots \\ u(0) \end{bmatrix} = \mathbf{x}_1 - \Phi^N \mathbf{x}_0.$$

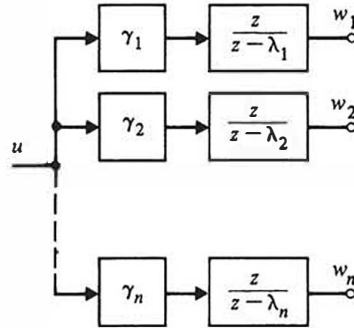
We have assumed that the dimension of the state, and hence the number of rows of the coefficient matrix of these equations, is n ; the number of columns is N . If N is less than n , we cannot possibly find a solution for every $\mathbf{x}_1$. If, on the other hand, N is greater than n , we will add a column $\Phi^n \Gamma$, and so on. But, by the Cayley-Hamilton theorem, (see Appendix C), Φ^n is a linear combination of lower powers of Φ , and the new columns add no new rank. Therefore we have a solution, and our system is controllable by definition **II** if and only if the rank of $\mathcal{C}$ is n , exactly the same condition as we found for pole assignment!

Our final definition is closest to the structural character of controllability.

III. The system (Φ, Γ) is controllable if every mode in Φ is connected to the control input.

Because of the generality of modes, we will treat only the case of systems for which Φ can be transformed to diagonal form. (The double-integrator model for the satellite does *not* qualify.) Suppose we have a diagonal Φ_λ matrix and

Figure 8.39
Block diagram for a
system with a diagonal
 Φ -matrix



corresponding input matrix Γ_λ with elements γ_i . Then the structure is as shown in Fig. 8.39. By the definition, the input must be connected to each mode so that no γ_i is zero. However, this is not enough if the roots λ_i are not distinct. Suppose, for instance, $\lambda_1 = \lambda_2$. Then the equations in the first two states are

$$\begin{aligned} w_1(k+1) &= \lambda_1 w_1(k) + \gamma_1 u, \\ w_2(k+1) &= \lambda_1 w_2(k) + \gamma_2 u. \end{aligned}$$

If we now define $\xi = \gamma_2 w_1 - \gamma_1 w_2$, the equation in ξ is

$$\gamma_2 w_1(k+1) - \gamma_1 w_2(k+1) = \lambda_1 \gamma_2 w_1(k) - \lambda_1 \gamma_1 w_2(k) + \gamma_1 \gamma_2 u - \gamma_1 \gamma_2 u,$$

which is the same as

$$\xi(k+1) = \lambda_1 \xi(k).$$

The point is that if two characteristic roots are equal *in a diagonal Φ_λ system with only one input*, we effectively have a hidden mode that is not connected to the control, and the system is not controllable. Therefore, even in this simple case, we have two conditions for controllability:

1. All characteristic values of Φ_λ are distinct, and
2. No element of Γ_λ is zero.

Now let us consider the controllability matrix of this diagonal system. By direct computation, we obtain

$$C = \begin{bmatrix} \gamma_1 & \gamma_1 \lambda_1 & \dots & \gamma_1 \lambda_1^{n-1} \\ \vdots & \gamma_2 \lambda_2 & \dots & \vdots \\ \gamma_n & \gamma_n \lambda_n & \dots & \gamma_n \lambda_n^{n-1} \end{bmatrix}$$

$$= \begin{bmatrix} \gamma_1 & 0 & & 0 \\ 0 & \gamma_2 & & \\ & & \ddots & \\ 0 & & & \gamma_n \end{bmatrix} \begin{bmatrix} 1 & \lambda_1 & \lambda_1^2 & \dots & \lambda_1^{n-1} \\ 1 & \lambda_2 & \lambda_2^2 & & \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \lambda_n & \lambda_n^2 & \dots & \lambda_n^{n-1} \end{bmatrix}.$$

The controllability matrix is a product of two terms, and $\mathcal{C}$ is nonsingular if and only if each factor is nonsingular. The first term has a determinant that is the product of the γ_i , and the second term is nonsingular if and only if the λ_i are distinct! So once again we find that our definition of controllability leads to the same test: The matrix $\mathcal{C}$ must be nonsingular. If $\mathcal{C}$ is nonsingular, then we can assign the system poles by state feedback, we can drive the state to any part of the space in finite time, and we know that every mode is connected to the input.¹⁹

As our final remark on the topic of controllability we present a test that is an alternative to testing the rank (or determinant) of $\mathcal{C}$. This is the Rosenbrock-Hautus-Popov (RHP) test [see Rosenbrock (1970), Kailath (1979)]. The system (Φ, Γ) is controllable if the system of equations

$$\mathbf{v}'[z\mathbf{I} - \Phi : \Gamma] = \mathbf{0}'$$

has only the trivial solution $\mathbf{v}' = \mathbf{0}'$, or, equivalently

$$\text{rank}\{z\mathbf{I} - \Phi : \Gamma\} = n,$$

or there is *no* nonzero $\mathbf{v}'$ such that

$$(i) \quad \mathbf{v}'\Phi = z\mathbf{v}', \quad (ii) \quad \mathbf{v}'\Gamma = 0.$$

This test is equivalent to the rank-of- $\mathcal{C}$ test. It is easy to show that if such a $\mathbf{v}$ exists, then $\mathcal{C}$ is singular. For if a nonzero $\mathbf{v}$ exists such that $\mathbf{v}'\Gamma = 0$ by (i), then, multiplying (i) by Γ on the right, we find

$$\mathbf{v}'\Phi\Gamma = z\mathbf{v}'\Gamma = 0.$$

Then, multiplying by $\Phi\Gamma$, we find

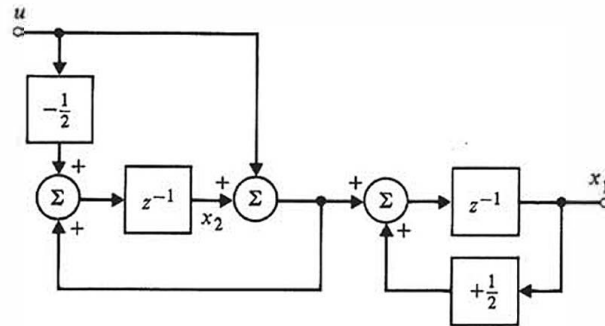
$$\mathbf{v}'\Phi^2\Gamma = z\mathbf{v}'\Phi\Gamma = 0,$$

and so on. Thus we derive $\mathbf{v}'\mathcal{C} = \mathbf{0}'$ has a nontrivial solution, $\mathcal{C}$ is singular, and the system is not controllable. To show that a nontrivial $\mathbf{v}'$ exists if $\mathcal{C}$ is singular requires a bit more work and is omitted. See Kailath (1979).

We have given two pictures of an uncontrollable system. Either the input is not connected to a dynamic part physically or else two parallel parts have identical characteristic roots. The engineer should be aware of the existence of a third simple situation, illustrated in Fig. 8.40. Here the problem is that the mode at $z = \frac{1}{2}$ appears to be connected to the input but is masked by the zero in the preceding member; the result is an uncontrollable system. First we will confirm

¹⁹ Of course, we showed this only for Φ that can be made diagonal. The result is true for general Φ .

Figure 8.40
Block diagram of a
simple uncontrollable
system



this allegation by computing the determinant of the controllability matrix. The system matrices are

$$\Phi = \begin{bmatrix} +\frac{1}{2} & 1 \\ 0 & 1 \end{bmatrix}, \quad \Gamma = \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix},$$

and

$$\mathcal{C} = [\Gamma \quad \Phi\Gamma] = \begin{bmatrix} 1 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix},$$

which is clearly singular. If we compute the transfer function from u to x_1 , we find

$$H(z) = \frac{z - \frac{1}{2}}{z - 1} \frac{1}{z - \frac{1}{2}} \quad (8.96)$$

$$= \frac{1}{z - 1}. \quad (8.97)$$

Because the natural mode at $z = \frac{1}{2}$ disappears, it is not connected to the input. Finally, if we consider the RHP test,

$$[z\mathbf{I} - \Phi \quad \Gamma] = \begin{bmatrix} z - \frac{1}{2} & -1 & 1 \\ 0 & z - 1 & \frac{1}{2} \end{bmatrix},$$

and let $z = \frac{1}{2}$, then we must test the rank of

$$\begin{bmatrix} 0 & -1 & 1 \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{bmatrix},$$

which is clearly less than two, which means, again, uncontrollable. In conclusion, we have three definitions of controllability: pole assignment, state reachability, and mode coupling to the input. The definitions are equivalent, and the tests for any of these properties are found in the rank of the controllability matrix or in the rank of the input system matrix $[z\mathbf{I} - \Phi \quad \Gamma]$.

observability

We have thus far discussed only controllability. The concept of observability

observability definitions

is parallel to that of controllability, and most of the results thus far discussed can be transferred to statements about observability by the simple expedient of substituting the transposes Φ^T for Φ and H^T for Γ . The result of these substitutions is a “dual” system. We have already seen an application of duality when we noticed that the conditions for the ability to select an observer gain L to give the state-error dynamics an arbitrary characteristic equation were that (Φ^T, H^T) must be controllable—and we were able to use the same Ackermann formula for estimator gain that we used for control gain. The other properties that are dual to controllability are

- OI.** The system (Φ, H) is observable if for any n th-order polynomial $\alpha_e(z)$, there exists an estimator gain L such that the characteristic equation of the state error of the estimator is $\alpha_e(z)$.
- OII.** The system (Φ, H) is observable if for any $x(0)$, there is a finite N such that $x(0)$ can be computed from observation of $y(0), y(1), \dots, y(N-1)$.
- OIII.** The system (Φ, H) is observable if every dynamic mode in Φ is connected to the output y via H .

We will consider the development of a test for observability according to definition **OII**. The system is described by²⁰

$$\begin{aligned}x(k+1) &= \Phi x(k), & x(0) &= x_0, \\y(k) &= Hx(k);\end{aligned}$$

and successive outputs from $k = 0$ are

$$\begin{aligned}y(0) &= Hx_0, \\y(1) &= Hx(1) = H\Phi x_0, \\y(2) &= Hx(2) = H\Phi x(1) = H\Phi^2 x_0, \\&\vdots \\y(N-1) &= H\Phi^{N-1} x_0.\end{aligned}$$

In matrix form, these equations are

$$\begin{bmatrix} y(0) \\ \vdots \\ y(N-1) \end{bmatrix} = \begin{bmatrix} H \\ H\Phi \\ \vdots \\ H\Phi^{N-1} \end{bmatrix} x_0.$$

²⁰ Clearly the input is irrelevant here if we assume that all values of $u(k)$ are available in the computation of x_0 . If some inputs, such as a disturbance w , are not available, we have a very different problem.

As we saw in the discussion of state controllability, new rows in these equations cannot be independent of previous rows if $N > n$ because of the Cayley-Hamilton theorem. Thus the test for observability is that the matrix

$$\mathcal{O} = \begin{bmatrix} \mathbf{H} \\ \mathbf{H}\Phi \\ \vdots \\ \mathbf{H}\Phi^{n-1} \end{bmatrix}$$

must be nonsingular. If we take the transpose of $\mathcal{O}$ and let $\mathbf{H}^T = \Gamma$ and $\Phi^T = \Phi$, then we find the controllability matrix of (Φ, Γ) , another manifestation of duality.

8.8 Summary

- For any controllable system (Φ, Γ) of order n , there exists a discrete full state feedback control law ($\mathbf{K}$) that will place the n closed-loop poles at arbitrary locations. `acker.m` or `place.m` using Φ, Γ perform this function.
- $\mathcal{C} = [\Gamma \quad \Phi\Gamma \quad \dots]$, the **controllability matrix**, must be of rank n , the order of the system, for the system to be controllable. `ctrb.m` performs this calculation.
- The general rule in selecting the desired pole locations is to move existing open-loop poles as little as possible in order to meet the system specifications.
- For any observable system $(\Phi, \mathbf{H})$ of order n , there exists a discrete estimator with gain $\mathbf{L}$ that will place the n estimator error equation poles at arbitrary locations. `acker.m` or `place.m` using $\Phi^T, \mathbf{H}^T$ calculates $\mathbf{L}$.
- $\mathcal{O} = [\mathbf{H} \quad \mathbf{H}\Phi \quad \dots]$, the **observability matrix**, must be of rank n , the order of the system, for the system to be observable. `obsf.m` performs this calculation.
- Feedback via $\mathbf{K}$ using the estimated state elements results in system poles that consist of the n control poles plus the n estimator poles.
- Estimator poles are usually selected to be approximately twice as fast as the controller poles in order for the response to be dominated by the control design. However, in order to smooth the effects of measurement noise, it is sometimes useful to select estimator poles as slow or slower than the control poles.
- Calculation of $\mathbf{N}_x$ and N_u via `refi.m` and their usage with the structure in Figs. 8.15 or 8.16, the **state command structure**, provides the best response to command inputs.
- **Integral control** is achieved by implementing the desired error integral and including this as part of an augmented plant model in calculating $\mathbf{K}$. The estimator is based on the non-augmented model. Integral control eliminates steady state errors due to command inputs and input disturbances.

- Disturbance estimation is accomplished by augmenting the model used in the estimator to include the unknown disturbance as a state element. The disturbance could be an unknown constant or a sinusoid with unknown magnitude and phase. The disturbance estimate can then be used in the control law to reject its effect, called **disturbance rejection** or to cause the system to track the disturbance with no error, called **disturbance following**.
- Delays in the actuator or sensor can be modeled in the estimator so that no estimation errors occur. **Sensor delays** will cause a delay in the response to a disturbance but there need not be any delay in the response to command inputs. However, **actuator delays** will cause a delayed response from command inputs and disturbances.

8.9 Problems

8.1 For the open-loop system

$$G(s) = \frac{y(s)}{u(s)} = \frac{1}{s^2 + 0.2s + 1},$$

- Find the discrete state-space representation assuming there is a ZOH and the sample period $T = 0.5$ sec.
- Find the full state digital feedback that provides equivalent s -plane poles at $\omega_n = 2$ rad/sec with $\zeta = 0.5$.
- Examine the response of the closed-loop system to an initial value of y and verify that the response is consistent with the desired poles.

8.2 For the open-loop system

$$\Phi = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \Gamma = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix},$$

compute $\mathbf{K}$ by hand so that the poles of the closed-loop system with full state feedback are at $z = 0.9 \pm j0.1$.

8.3 For the open-loop system

$$G(s) = \frac{y(s)}{u(s)} = \frac{1}{s^2(s+2)},$$

- Find the discrete state-space model, assuming there is a ZOH and the sample rate is $T = 100$ msec.
- Pick poles so that the settling time $t_s < 1$ sec and find the $\mathbf{K}$ that will produce those poles with full state feedback.
- Verify that t_s is satisfied by plotting the response to an initial value of y .

8.4 For the system in Problem 8.2, find the estimator equations and the value of the gain $\mathbf{L}$ by hand so that $z_{\text{des}} = 0.6 \pm j0.3$ for

- a predictor estimator,
- a current estimator, and
- a reduced-order estimator ($z_{\text{des}} = 0.6$).

8.5 For the system in Problem 8.3,

- (a) Find the predictor estimator equations and the value of the gain L_p so that the estimator $t_s < 0.5$ sec.
- (b) Verify that t_s is satisfied by plotting the response of $\hat{x}_1 (= \hat{y})$ to an initial value.

8.6 For the open-loop system

$$G(s) = \frac{y(s)}{u(s)} = \frac{1}{s^2}$$

preceded by a ZOH and a sample rate of 20 Hz,

- (a) Find the feedback gain K so that the control poles have an equivalent s -plane $\omega_n = 10$ rad/sec and $\zeta = 0.7$.
 - (b) Find the estimator gain L_p so that the estimator poles have an equivalent s -plane $\omega_n = 20$ rad/sec and $\zeta = 0.7$.
 - (c) Determine the discrete transfer function of the compensation.
 - (d) Design a lead compensation using transform techniques so that the equivalent s -plane natural frequency $\omega_n \cong 10$ rad/sec and $\zeta \cong 0.7$. Use either root locus or frequency response.
 - (e) Compare the compensation transfer functions from (c) and (d) and discuss the differences.
- 8.7 For the system in Problem 8.6, design the controller and estimator so that the closed-loop unit step response to a command input has a rise time $t_r < 200$ msec and an overshoot $M_p < 15\%$ when using:
- (a) the state command structure,
 - (b) the output error command structure.

For both cases, check that the specifications are met by plotting the step response.

8.8 For the open-loop system

$$G(s) = \frac{y(s)}{u(s)} = \frac{1}{s^2(s^2 + 400)},$$

design the controller and estimator so that the closed-loop unit step response to a command input has a rise time $t_r < 200$ msec and an overshoot $M_p < 15\%$ when using the state command structure. Check that the specifications are met by plotting the step response. $T = 30$ ms.

8.9 Compute $G(z)$ from Eq. (4.64) for

$$\Phi = \begin{bmatrix} -a_1 & -a_2 & -a_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \Gamma = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{H} = [b_1 \quad b_2 \quad b_3].$$

Why is this form for Φ and Γ called control canonical form?

8.10 (a) For

$$\Phi = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \quad \Gamma = \begin{bmatrix} T^2/2 \\ T \end{bmatrix},$$

find a transform matrix T so that if $\mathbf{x} = T\mathbf{w}$, then the equations in $\mathbf{w}$ will be in control canonical form.

- (b) Compute $\mathbf{K}_w$, the gain, such that if $u = -\mathbf{K}_w \mathbf{w}$, the characteristic equation will be $\alpha_c(z) = z^2 - 1.6z + 0.7$.
- (c) Use $\mathbf{T}$ from part (a) to compute $\mathbf{K}_x$, the gain in the x -states.
- 8.11 (a) Show that the equations for the current estimator can be written in standard state form

$$\xi_{k+1} = \mathbf{A}\xi_k + \mathbf{B}y_k, \quad u = \mathbf{C}\xi_k + D y_k,$$

where $\xi_k = \hat{x}_k - \mathbf{L}_c y_k$, $\mathbf{A} = (\mathbf{I} - \mathbf{L}_c \mathbf{H})(\Phi - \Gamma \mathbf{K})$, $\mathbf{B} = \mathbf{A} \mathbf{L}_c$, $\mathbf{C} = -\mathbf{K}$, and $D = -\mathbf{K} \mathbf{L}_c$.

- (b) Use the results of Eq. (4.65) to show that the controller based on a current estimator always has a zero at $z = 0$ for any choice of control law $\mathbf{K}$ or estimator law $\mathbf{L}_c$.
- 8.12 For the open-loop system

$$\Phi = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad \Gamma = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix},$$

check the observability for:

- (a) $\mathbf{H} = [0 \quad 1]$.
- (b) $\mathbf{H} = [1 \quad 0]$.
- (c) Rationalize your results to (a) and (b), stating why the observability or lack of it occurred.
- 8.13 Design the antenna in Appendix A.2 by state-variable pole assignment.
- (a) Write the equations in state form with $x_1 = y$ and $x_2 = \dot{y}$. Give the matrices $\mathbf{F}$, $\mathbf{G}$, and $\mathbf{H}$. Let $a = 0.1$.
- (b) Let $T = 1$ and design $\mathbf{K}$ for equivalent poles at $s = -1/2 \pm j(\sqrt{3}/2)$. Plot the step response of the resulting design.
- (c) Design a prediction estimator with $\mathbf{L}_p$ selected so that $\alpha_e(z) = z^2$; that is, both poles are at the origin.
- (d) Use the estimated states for computing the control and introduce the reference input so as to leave the state estimate undisturbed. Plot the unit step response from this reference input and from a wind gust (step) disturbance that acts on the antenna just like the control force (but not on the estimator).
- (e) Plot the root locus of the closed-loop system with respect to the plant gain and mark the locations of the closed-loop poles.
- 8.14 In Problem 7.8 we described an experiment in magnetic levitation described by the equations

$$\ddot{x} = 1000x + 20u.$$

Let the sampling time, T , be 10 msec.

- (a) Use pole placement to design this system to meet the specifications that settling time is less than 0.25 sec and overshoot to an initial offset in x is less than 20%.
- (b) Design a reduced-order estimator for $\dot{x}$ for this system such that the error-settling time will be less than 0.08 sec.
- (c) Plot step responses of x , $\dot{x}$, and u for an initial x displacement.
- (d) Plot the root locus for changes in the plant gain and mark the design pole locations.

- (e) Introduce a command reference with feedforward so that the estimate of $\dot{x}$ is not forced by r . Measure or compute the frequency response from r to system error $r - x$ and give the highest frequency for which the error amplitude is less than 20% of the command amplitude.

8.15 Derive Eq. (8.63) from Eq. (8.57).

8.16 For the open-loop system

$$G(z) = \frac{y(z)}{u(z)} = \frac{y(z)}{w(z)} = \frac{0.1185(z + 0.9669)}{z^2 - 1.6718z + 0.9048},$$

- Find the control feedback $\mathbf{K}$ and estimator gain $\mathbf{L}_p$ that will place control poles at $z = 0.8 \pm j0.2$ and estimator poles at $z = 0.6 \pm j0.3$.
- Plot the response of $y(k)$ for a unit step input (r) command using the state command structure. Would there be a steady-state error if $N_u = 0$?
- Determine what the steady state value of $y(k)$ would be if there was an input disturbance, w .
- Determine an integral control gain and show the system block diagram with the integral control term included. Set the extra control pole at $z = 0.9$.
- Demonstrate that the system will have no steady state error for a disturbance, w , or an input command, u , even when $N_u = 0$.

8.17 For the open-loop system

$$\Phi = \begin{bmatrix} 0.8815 & 0.4562 \\ -0.4562 & 0.7903 \end{bmatrix}, \quad \Gamma = \Gamma_1 = \begin{bmatrix} 0.1185 \\ 0.4562 \end{bmatrix}, \quad \mathbf{H} = [1 \quad 0],$$

- Find the control feedback $\mathbf{K}$ and estimator gain $\mathbf{L}_p$ that will place control poles at $z = 0.6 \pm j0.3$ and estimator poles at $z = 0.3 \pm j0.3$.
- Plot the response of $y(k)$ for a unit step of the disturbance (w).
- Determine an integral control gain and show the system block diagram with the integral control term included. Set the extra control pole at $z = 0.9$.
- Demonstrate that the system will have no steady-state error for a disturbance, w .

8.18 For the open-loop system from Problem 8.17

- Assuming w takes on some unknown but constant value, construct an estimator that includes an estimate of that disturbance. Place poles of the system as in Problem 8.17, except place the extra estimator pole at $z = 0.9$. Determine values of $\mathbf{K}$ and $\mathbf{L}_p$ and sketch the block diagram showing how the various quantities are used in the control. Include the command input r in the diagram using the state command structure.
- Plot the response of y and $\tilde{w}$ to a unit step in w with $r \equiv 0$. State whether the responses meet your expectations.
- Plot the response of y and $\tilde{w}$ to a unit step in r with $w \equiv 0$. State whether the responses meet your expectations.

8.19 A disk drive read head has the open-loop transfer function

$$G(s) = \frac{y(s)}{u(s)} = \frac{y(s)}{w(s)} = \frac{1000\omega_r^2}{s^2(s^2 + 2\zeta_r\omega_r s + \omega_r^2)}$$

where $\omega_r = 6000$ rad/sec and $\zeta_r = 0.02$.

- Design a digital compensation so that there is no steady state error to an input command nor to a constant disturbance w . The plant gain of 1000 is not known precisely, so it is not acceptable to assume the steady state error due to input commands can be eliminated via feedforward with an N_u term. The specifications are that the rise time t_r must be less than 2 msec and the overshoot $M_p < 10\%$. Use a sample rate of 3 kHz.
- The disk spins at 3000 rpm. There is typically a small offset between the center of the circular tracks on the disk and the center of rotation, thus producing a wobble in the location of the tracks that should be followed by the read head. Determine the track following error due to this wobble. Express the answer as a percentage of the wobble magnitude.
- Embellish your design from (a) so that the error due to the wobble is eliminated as best you can. Plot the frequency response of the tracking error (e in Fig. 8.32) where the input to the system is the track wobble. Mark the frequency that represents the spin rpm.

8.20 A pendulum with a torquer at its hinge is described by

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u + \begin{bmatrix} 0 \\ 1 \end{bmatrix} w,$$

$$y = [1 \quad 0] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + b$$

where $\mathbf{x}^T = [\theta \quad \dot{\theta}]$, θ = angle of the pendulum from vertical, u = torque, w = torque bias, and b = measurement bias. Answer the questions below assuming the output is sampled with $T = 100$ msec and the control $(u + w)$ is applied through a ZOH.

- With no torque bias ($w = 0$), augment the system model so that the measurement bias is a state element. Is this system observable?
 - With no measurement bias ($b = 0$), augment the system model so that the torque bias is a state element. Is this system observable?
 - Augment the system model so that both biases are state elements. Is this system observable?
- 8.21 Design a controller (control plus estimator) for the same system as in Example 8.12, except add a delay of two sample periods between the system output and when the measurement is available to the computer.
- Compute $\mathbf{K}$ for $z_{des} = 0.8 \pm j0.25, 0, 0$.
 - Is the system observable? Check for predictor and current estimators.
 - Compute $\mathbf{L}_p$ with $z_{des} = 0.4 \pm j0.4, 0, 0$.
 - Compute $\mathbf{L}_c$ with z_{des} as in part (c) and with $z_{des} = 0.4 \pm j0.4, 0.1, 0$.
 - Plot the unit step response to an input command using the predictor estimator showing the plant output as well as the delayed sensor output.

8.22 Determine the state-command input structure for a feedback system with the Type 0 plant

$$G(s) = \frac{10}{s^2 + 0.18s + 9}.$$

- Convert the system to discrete state-space form with $T = 0.1$ sec.

- (b) Find $\mathbf{K}$ to obtain equivalent s -plane control poles at $\zeta = 0.5$ and $\omega_n = 4$ rad/sec.
 - (c) Find $\mathbf{L}_p$ to obtain equivalent s -plane estimator poles at $\zeta = 0.5$ and $\omega_n = 8$ rad/sec.
 - (d) Determine $\mathbf{N}_u$ and $\mathbf{N}_x$, then sketch a block diagram specifying the controller equations including the reference input.
 - (e) Do you expect there will be a steady-state error for this system for a step input?
 - (f) Plot the step response and confirm your answer to (e).
- 8.23 Repeat the design of the controller (control and estimator) for Examples 8.4 and 8.8, but place all four desired poles at $z = 0$. (This is often referred to as a *finite settling time* or *deadbeat* design because the system will settle in a finite number of sample periods.)
- (a) Using your $\mathbf{K}$ and $\mathbf{L}_p$, determine the time response of the system to a unit-step input using the state-command input structure.
 - (b) Determine the compensation in transfer function form, $D_p(z)$, and construct a root locus vs. the dc gain of $D_p(z)$. Mark the roots where the gain corresponds to the values computed for $\mathbf{K}$ and $\mathbf{L}_p$.
- 8.24 The double mass-spring device described in Appendix A.4 is representative of many devices that have some structural resonances. Placing the sensor so that it measures y is called the *colocated* case, whereas placing it so that it measures d is called the *noncolocated* case. Often, the designer is not aware initially that a resonance exists in the system, a situation that is addressed by this problem.

For $M = 20$ kg, $m = 1$ kg, $k = 25$ N/m, and $b = 0.2$ N-sec/m, we obtain a resonance frequency of 5 rad/sec with a damping ratio, $\zeta = 0.02$.

- (a) To represent the case where the designer did not know about the resonance, assume the coupling is rigid, that is, k is infinite. The transfer function is then

$$G_1(s) = \frac{d(s)}{u(s)} = \frac{y(s)}{u(s)} = \frac{1}{(m + M)s^2}.$$

Design a digital controller ($\mathbf{K}$ and $\mathbf{L}_p$) with $T = 200$ msec, control poles at $z = 0.75 \pm j0.2$, and estimator poles at $z = 0.3 \pm j0.3$. Verify by simulation that it provides a response to a unit-step command using the state-command input structure that is consistent with the selected poles.

- (b) Use the controller ($\mathbf{K}$ and $\mathbf{L}_p$) obtained in part (a) in a simulation of the system where the infinite spring is replaced with the flexible one and the output is d , that is, a fourth-order plant with second-order controller. Examine the response and compare it qualitatively with an analysis of the closed-loop roots of this combined system.
- (c) Repeat part (b), but replace the plant output, d , with y .
- (d) Analyze where the roots of the system would be if you measured y and $\dot{y}$ directly (no estimator) and fed them back using your $\mathbf{K}$ from part (a).
- (e) Design a fourth-order controller with control poles at $z = 0.75 \pm j0.2$, $0.4 \pm j0.6$, and estimator poles at $z = 0.3 \pm j0.3$, $0 \pm j0.4$ with d as the measurement. Again, verify by simulation that it provides the correct response to a unit-step command using the state-command input structure.
- (f) Plot the frequency response of the compensation (control plus estimator) from part (e). State why you think this kind of compensation is usually referred to as a *notch filter*.

- (g) Plot the z -plane root locus of the system (plant plus controller) and comment on the sensitivity of this design to changes in the overall loop gain.

8.25 A heat exchanger²² has the transfer function

$$G(s) = \frac{e^{-5s}}{(10s + 1)(60s + 1)}$$

where the delay is due to the sensor.

- (a) Write state equations for this system.
- (b) Compute the zero-order-hold model with a sample period of 5 sec.
- (c) Design the compensation including the command input with the control poles at $0.8 \pm j0.25$ and the estimator poles at $0.4 \pm j0.4, 0$.
- (d) Compute the step response to a reference input step and to a disturbance input at the control. Verify that there is no delay for the command input.

²² See Franklin, Powell, and Emami (2019), Example 2.18.

• 9 •

Multivariable and Optimal Control

A Perspective on Multivariable and Optimal Control

The control-design procedures described in Chapters 7 and 8 were applied to systems with a single input and single output (SISO). The transfer-function approach in Chapter 7 is best suited to SISO systems; the state-space methods of Chapter 8 were limited to SISO in order to simplify the procedures. In fact, if we try to apply the pole-placement approach of Chapter 8 to a multivariable (multi-input, multi-output, or MIMO) system, we find that the gains, $\mathbf{K}$ or $\mathbf{L}$, are not uniquely determined by the resulting equations. Therefore, a design approach is required which intelligently uses this extra freedom for MIMO systems. In addition, we saw in Example 8.3 that the selection of desired pole locations for SISO systems can be tricky business. Some sort of systematic guidance for the selection of control and estimator pole locations seems highly desirable. The material in this chapter provides a tool that meets both these needs.

The subject of this chapter is the use of optimal control techniques as a tool in the design of control systems. It is important that the designer have no illusions that some true “optimal” design is being achieved; rather, the idea is to transfer the designer’s iteration on pole locations as used in Chapter 8, or compensation parameters as used in Chapter 7, to iterations on elements in a cost function, $\mathcal{J}$. The method will determine the control law that minimizes $\mathcal{J}$, but because the parameters in $\mathcal{J}$ are arbitrarily selected, the design is at best only partially optimal. However, these designs will achieve some compromise between the use of control effort and the speed of response and will guarantee a stable system, no small feat. Therefore, each iteration on the parameters in $\mathcal{J}$ produces a candidate design that should be evaluated in the light of the design specifications.

Chapter Overview

The chapter starts out in Section 9.1 by discussing some of the steps that might be taken in order to convert a MIMO system into a SISO one. Although this cannot

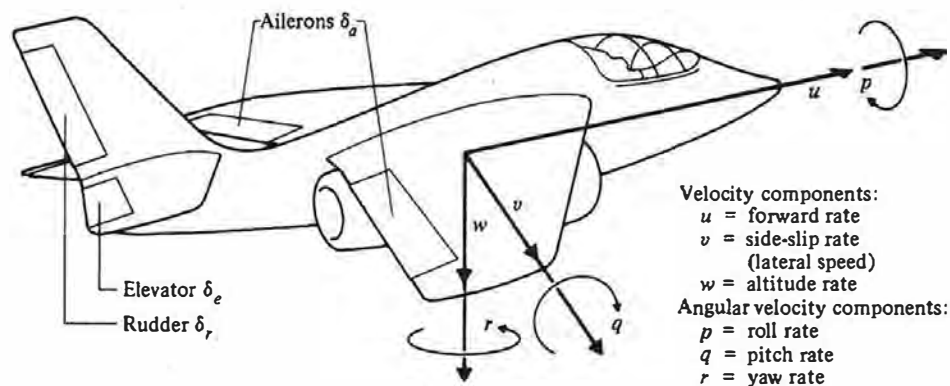
always be done, it can help to clarify the key control issues in a complex system so that the later optimization of the whole system will be better understood. Section 9.2 derives the time-varying optimal control solution that results directly from the optimal control problem statement. Section 9.3 shows how to find the steady-state value of the optimal feedback gain that is significantly easier to implement and is the one typically used in control implementations. Section 9.4 derives the companion optimal estimation problem. As for the control case, the time-varying gain solution is found first, then we find the steady-state gain case that is typically the one implemented. The final section, 9.5, shows how to use these results in the design of MIMO control systems.

9.1 Decoupling

The first step in any multivariable design should be an attempt either to find an approximate model consisting of two or more single input-output models or else to decouple the control gain matrix $\mathbf{K}$ and the estimator gain matrix $\mathbf{L}$. This step will give better physical insight into the important feedback variables and can lead to a plant description that is substantially simpler for design purposes and yet yields no significant degradation from an analysis based on the full multivariable system.

For example, the linearized equations of motion of an aircraft (Fig. 9.1) are of eighth order but are almost always separated into two fourth-order sets representing longitudinal motion (w, u, q) and lateral motion (p, r, v). The elevator

Figure 9.1
Schematic of an aircraft showing variable definitions



control surfaces affect longitudinal motion; the aileron and rudder primarily affect lateral motion. Although there is a small amount of coupling of lateral motion into longitudinal motion, this is ignored with no serious consequences when the control, or “stability-augmentation,” systems are designed independently for the two fourth-order systems.

Further decoupling of the equations is also possible.

◆ Example 9.1 Control Decoupling

Decouple the lateral aircraft fourth-order equations into two second-order equations and show how to design the control.

Solution. The aircraft lateral equations are multivariable and of the form

$$\mathbf{x}(k+1) = \Phi\mathbf{x}(k) + \Gamma\mathbf{u}(k), \quad (9.1)$$

where

$$\mathbf{u} = \begin{bmatrix} \delta_r \\ \delta_a \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} v \\ r \\ \phi_p \\ p \end{bmatrix}, \quad \text{and} \quad p = \dot{\phi}_p.$$

A control law of the standard form

$$\begin{bmatrix} \delta_r \\ \delta_a \end{bmatrix} = - \begin{bmatrix} K_{11} & K_{12} & K_{13} & K_{14} \\ K_{21} & K_{22} & K_{23} & K_{24} \end{bmatrix} \begin{bmatrix} v \\ r \\ \phi_p \\ p \end{bmatrix} \quad (9.2)$$

shows that there are eight elements in the gain matrix to be selected, and the specification of four closed-loop roots clearly causes the problem to be underdetermined and will leave many possible values of $\mathbf{K}$ that will meet the specifications. This shows that the pole-placement approach to multivariable system design poses difficulties and additional criteria need to be introduced.

A decoupling that removes the ambiguity is to restrict the control law to

$$\begin{bmatrix} \delta_r \\ \delta_a \end{bmatrix} = - \begin{bmatrix} K_{11} & K_{12} & 0 & 0 \\ 0 & 0 & K_{23} & K_{24} \end{bmatrix} \begin{bmatrix} v \\ r \\ \phi_p \\ p \end{bmatrix}. \quad (9.3)$$

This makes good physical sense because the rudder primarily yaws the aircraft about a vertical axis (r -motion), thus directly causing sideslip (v), and the ailerons primarily roll the aircraft about an axis through the nose, thus causing changes in the roll angle, ϕ_p , and the roll rate, p . Given an achievable set of desired pole locations, there are unique values of the four nonzero components of $\mathbf{K}$; however, the governing equations cannot be cast in the same form as in Eq. (8.5) and therefore can be difficult to solve.

A further decoupling that would permit an easy gain calculation is to assume that Eq. (9.1) is of the form

$$\begin{bmatrix} v \\ r \\ \phi_p \\ p \end{bmatrix}_{k+1} = \begin{bmatrix} \phi_{11} & \phi_{12} & 0 & 0 \\ \phi_{21} & \phi_{22} & 0 & 0 \\ 0 & 0 & \phi_{33} & \phi_{34} \\ 0 & 0 & \phi_{43} & \phi_{44} \end{bmatrix} \begin{bmatrix} v \\ r \\ \phi_p \\ p \end{bmatrix}_k + \begin{bmatrix} \Gamma_{11} & 0 \\ \Gamma_{21} & 0 \\ 0 & \Gamma_{32} \\ 0 & \Gamma_{42} \end{bmatrix} \begin{bmatrix} \delta_r \\ \delta_a \end{bmatrix}_k \quad (9.4)$$

and that the control law is given by Eq. (9.3).

This makes some physical sense but ignores important coupling between the two aircraft modes. It does, however, decouple the system into second-order systems for which the methods of Chapter 8 can be applied directly to obtain the gains. The resulting closed-loop characteristic roots of the full lateral equations can be checked by calculating the eigenvalues of the closed loop matrix: (see eig.m in MATLAB)

$$\Phi_{\text{closed loop}} = \begin{bmatrix} \phi_{11} - \Gamma_{11}K_{11} & \phi_{12} - \Gamma_{11}K_{12} & \phi_{13} - \Gamma_{12}K_{23} & \phi_{14} - \Gamma_{12}K_{24} \\ \phi_{21} - \Gamma_{21}K_{11} & \phi_{22} - \Gamma_{21}K_{12} & \phi_{23} - \Gamma_{22}K_{23} & \phi_{24} - \Gamma_{22}K_{24} \\ \phi_{31} - \Gamma_{31}K_{11} & \phi_{32} - \Gamma_{31}K_{12} & \phi_{33} - \Gamma_{32}K_{23} & \phi_{34} - \Gamma_{32}K_{24} \\ \phi_{41} - \Gamma_{41}K_{11} & \phi_{42} - \Gamma_{41}K_{12} & \phi_{43} - \Gamma_{42}K_{23} & \phi_{44} - \Gamma_{42}K_{24} \end{bmatrix}, \quad (9.5)$$

which results from combining Eq. (9.3) and (9.4).

If the plant coupling that was ignored in the gain computation is important, the roots obtained from Eq. (9.5) will differ from those used to compute the gains using Eqs. (9.3) and (9.4). In many cases, the method will be accurate enough and one need look no further. In other cases, one could revise the “desired” root locations and iterate until the correct roots from Eq. (9.5) are satisfactory or else turn to the methods of optimal control to be described in the following sections.

The same ideas apply equally well to the decoupling of the estimator into SISO parts.

◆ Example 9.2 Estimator Decoupling

Decouple the estimator for the inverted pendulum on a motorized cart (Fig. 9.2).

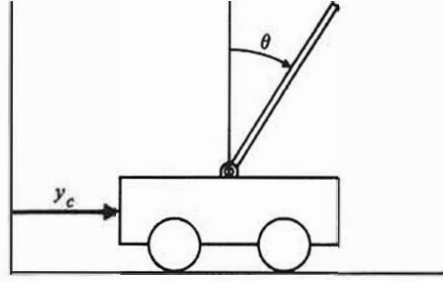
Solution. The equations of motion can be written as

$$\begin{bmatrix} \mathbf{x}_c \\ \mathbf{x}_s \end{bmatrix}_{k+1} = \begin{bmatrix} \phi_{cc} & \phi_{cs} \\ \phi_{sc} & \phi_{ss} \end{bmatrix} \begin{bmatrix} \mathbf{x}_c \\ \mathbf{x}_s \end{bmatrix}_k + \begin{bmatrix} \Gamma_c \\ \Gamma_s \end{bmatrix} u(k), \quad (9.6)$$

where

$$\begin{aligned} \mathbf{x}_c &= \begin{bmatrix} y_c \\ \dot{y}_c \end{bmatrix} \text{ cart position and velocity,} \\ \mathbf{x}_s &= \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} \text{ stick angle and angular rate,} \end{aligned}$$

Figure 9.2
Hinged stick and
motorized cart



and the available measurements are

$$\begin{bmatrix} y \\ \theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} y_c \\ \dot{y}_c \\ \theta \\ \dot{\theta} \end{bmatrix}. \quad (9.7)$$

The stick pictured in Fig. 9.2 is substantially lighter than the cart. This means that stick motion has a small dynamic effect on cart motion, which in turn implies that $\phi_{cs} \cong 0$. This does not imply that $\phi_{sc} = 0$; in fact, cart motion is the mechanism for influencing stick motion and hence stabilizing it.

An estimator for the system described by Eq. (9.6) and Eq. (9.7) requires the determination of eight elements of an estimator gain matrix, $\mathbf{L}$. Hence, specifying the four estimator roots and using the methods of Chapter 8 would not determine this $\mathbf{L}$ uniquely—another example of an underdetermined system caused by the multivariable nature of the problem.

But because we can assume that $\phi_{cs} = 0$, the cart equation in Eq. (9.6) uncouples from the stick equation, and we simply design an estimator for

$$\begin{aligned} \mathbf{x}_c(k+1) &= \phi_{cc} \mathbf{x}_c(k) + \Gamma_c u, \\ y_c &= [1 \quad 0] \begin{bmatrix} y_c \\ \dot{y}_c \end{bmatrix}, \end{aligned} \quad (9.8)$$

which can be done with the methods described in Chapter 8. There is one-way coupling into the stick equation, but this just acts like an additional control input and can be ignored in the calculation of the stick estimator gain matrix, $\mathbf{L}_s$, using the pole-placement methods of Chapter 8. However, the ϕ_{sc} coupling should not be ignored in the estimator equations, and there is no reason to ignore the weak coupling, ϕ_{cs} . The final (predictor) estimator would be of the form

$$\begin{aligned} \bar{\mathbf{x}}_c(k+1) &= \phi_{cc} \bar{\mathbf{x}}_c(k) + \phi_{cs} \bar{\mathbf{x}}_s(k) + \Gamma_c u(k) + \mathbf{L}_c (y_c(k) - \bar{y}_c(k)), \\ \bar{\mathbf{x}}_s(k+1) &= \phi_{sc} \bar{\mathbf{x}}_c(k) + \phi_{ss} \bar{\mathbf{x}}_s(k) + \Gamma_s u(k) + \mathbf{L}_s (\theta(k) - \bar{\theta}(k)), \end{aligned}$$

where $\mathbf{L}_c$ and $\mathbf{L}_s$ are both 2×1 matrices.

Even without the very weak one-way coupling in ϕ_{cs} that was obvious for this example, one could assume this to be the case, then check the resulting full-system characteristic roots using a method similar to the previous airplane example. Note that ignoring the coupling only causes approximations in the gain-matrix calculation and thus the root locations. There is no

approximation in the system model used in the estimator; therefore, the estimation errors will still approach zero for stable estimator roots.

In short, it is often useful to apply your knowledge of the physical aspects of the system at hand to break the design into simpler and more tractable subsets. With luck, the whole job can be finished this way. At worst, insight will be gained that will aid in the design procedures to follow and in the implementation and checkout of the control system.

9.2 Time-Varying Optimal Control

Optimal control methods are attractive because they handle MIMO systems easily and aid in the selection of the desired pole locations for SISO systems. They also allow the designer to determine many good candidate values of the feedback gain, $\mathbf{K}$, using very efficient computation tools. We will develop the time-varying optimal control solution first and then reduce it to a steady-state solution in the following section. The result amounts to another method of computing $\mathbf{K}$ in the control law Eq. (8.5)

$$\mathbf{u} = -\mathbf{K}\mathbf{x} \quad (9.9)$$

that was used in Chapter 8 and illustrated by Eq. (9.2) in Example 9.1.

Given a discrete plant

$$\mathbf{x}(k+1) = \Phi\mathbf{x}(k) + \Gamma\mathbf{u}(k), \quad (9.10)$$

cost function

we wish to pick $\mathbf{u}(k)$ so that a **cost function**

$$\mathcal{J} = \frac{1}{2} \sum_{k=0}^N [\mathbf{x}^T(k)\mathbf{Q}_1\mathbf{x}(k) + \mathbf{u}^T(k)\mathbf{Q}_2\mathbf{u}(k)] \quad (9.11)$$

is minimized. $\mathbf{Q}_1$ and $\mathbf{Q}_2$ are symmetric weighting matrices to be selected by the designer, who bases the choice on the relative importance of the various states and controls. Some weight will almost always be selected for the control ($|\mathbf{Q}_2| \neq 0$); otherwise the solution will include large components in the control gains, and the states would be driven to zero at a very fast rate, which could saturate the actuator device.¹ The $\mathbf{Q}$'s must also be **nonnegative definite**,² which is most

¹ If the sampling rate, T , is long, however, a control that moves the state along as rapidly as possible might be feasible. Such controls are called "dead-beat" because they beat the state to a dead stop in at most n steps. They correspond to placement of all poles at $z = 0$. See Problem 8.23.

² Matrix equivalent of a nonnegative number; it ensures that $\mathbf{x}^T\mathbf{Q}_1\mathbf{x}$ and $\mathbf{u}^T\mathbf{Q}_2\mathbf{u}$ are nonnegative for all possible $\mathbf{x}$ and $\mathbf{u}$.

easily accomplished by picking the $\mathbf{Q}$'s to be diagonal with all diagonal elements positive or zero.

Another way of stating the problem given by Eqs. (9.10) and (9.11) is that we wish to minimize

$$\mathcal{J} = \frac{1}{2} \sum_{k=0}^N [\mathbf{x}^T(k) \mathbf{Q}_1 \mathbf{x}(k) + \mathbf{u}^T(k) \mathbf{Q}_2 \mathbf{u}(k)] \quad [9.11]$$

subject to the constraint that

$$-\mathbf{x}(k+1) + \Phi \mathbf{x}(k) + \Gamma \mathbf{u}(k) = 0, \quad k = 0, 1, \dots, N. \quad [9.10]$$

Lagrange multipliers

This is a standard constrained-minima problem which can be solved using the method of **Lagrange multipliers**. There will be one Lagrange multiplier vector, which we will call $\boldsymbol{\lambda}(k+1)$, for each value of k . The procedure is to rewrite Eqs. (9.10) and (9.11) as

$$\begin{aligned} \mathcal{J}' = \sum_{k=0}^N & \left[\frac{1}{2} \mathbf{x}^T(k) \mathbf{Q}_1 \mathbf{x}(k) + \frac{1}{2} \mathbf{u}^T(k) \mathbf{Q}_2 \mathbf{u}(k) \right. \\ & \left. + \boldsymbol{\lambda}^T(k+1) (-\mathbf{x}(k+1) + \Phi \mathbf{x}(k) + \Gamma \mathbf{u}(k)) \right], \end{aligned} \quad (9.12)$$

and find the minimum of $\mathcal{J}'$ with respect to $\mathbf{x}(k)$, $\mathbf{u}(k)$, and $\boldsymbol{\lambda}(k)$. Note that for an optimal $\mathbf{u}(k)$ that obeys Eq. (9.10), the two cost functions, $\mathcal{J}'$ and $\mathcal{J}$, are identical in magnitude. The index on $\boldsymbol{\lambda}$ is arbitrary conceptually, but we let it be $k+1$ because this choice will yield a particularly easy form of the equations later on.

Proceeding with the minimization leads to

$$\frac{\partial \mathcal{J}'}{\partial \mathbf{u}(k)} = \mathbf{u}^T(k) \mathbf{Q}_2 + \boldsymbol{\lambda}^T(k+1) \Gamma = 0, \quad \text{control equations,} \quad (9.13)$$

$$\frac{\partial \mathcal{J}'}{\partial \boldsymbol{\lambda}(k+1)} = -\mathbf{x}(k+1) + \Phi \mathbf{x}(k) + \Gamma \mathbf{u}(k) = 0, \quad \text{state equations, and} \quad [9.10]$$

$$\frac{\partial \mathcal{J}'}{\partial \mathbf{x}(k)} = \mathbf{x}^T(k) \mathbf{Q}_1 - \boldsymbol{\lambda}^T(k) + \boldsymbol{\lambda}^T(k+1) \Phi = 0, \quad \text{adjoint equations.} \quad (9.14)$$

adjoint equations

The last set of the equations, the **adjoint equations**, can be written as the backward difference equation

$$\boldsymbol{\lambda}(k) = \Phi^T \boldsymbol{\lambda}(k+1) + \mathbf{Q}_1 \mathbf{x}(k). \quad (9.15)$$

Restating the results in more convenient forms, we have from Eq. (9.10)

$$\mathbf{x}(k+1) = \Phi \mathbf{x}(k) + \Gamma \mathbf{u}(k), \quad (9.16)$$

where, using Eq. (9.13)

$$\mathbf{u}(k) = -\mathbf{Q}_2^{-1} \Gamma^T \boldsymbol{\lambda}(k+1), \quad (9.17)$$

and from Eq. (9.15) we can describe $\lambda(k+1)$ in the forward difference equation form

$$\lambda(k+1) = \Phi^{-T} \lambda(k) - \Phi^{-T} Q_1 x(k). \quad (9.18)$$

Equations (9.16), (9.17), and either (9.15) or (9.18) are a set of coupled difference equations defining the optimal solution of $x(k)$, $\lambda(k)$, and $u(k)$, provided the initial (or final) conditions are known. The initial conditions on $x(k)$ must be given; however, usually $\lambda(0)$ would not be known, and we are led to the endpoint to establish a boundary condition for λ . From Eq. (9.11) we see that $u(N)$ should be zero in order to minimize J because $u(N)$ has no effect on $x(N)$ [see Eq. (9.10)]. Thus Eq. (9.13) suggests that $\lambda(N+1) = 0$, and Eq. (9.14) thus shows that a suitable boundary condition is

$$\lambda(N) = Q_1 x(N). \quad (9.19)$$

A set of equations describing the solution to the optimal control problem is now completely specified. It consists of the two difference equations (9.16) and (9.15) with u given by Eq. (9.17), the final condition on λ given by Eq. (9.19), and the initial condition on x would be given in the problem statement. The solution to this two-point boundary-value problem is not easy.

sweep method

One method, called the **sweep method** by Bryson and Ho (1975), is to assume

$$\lambda(k) = S(k)x(k). \quad (9.20)$$

This definition allows the transformation of the two-point boundary-value problem in x and λ to one in S with a single-point boundary condition. With the definition Eq. (9.20), the control Eq. (9.13) becomes

$$\begin{aligned} Q_2 u(k) &= -\Gamma^T S(k+1)x(k+1) \\ &= -\Gamma^T S(k+1)(\Phi x(k) + \Gamma u(k)). \end{aligned}$$

Solving for $u(k)$, we obtain

$$\begin{aligned} u(k) &= -(Q_2 + \Gamma^T S(k+1)\Gamma)^{-1} \Gamma^T S(k+1)\Phi x(k) \\ &= -R^{-1} \Gamma^T S(k+1)\Phi x(k). \end{aligned} \quad (9.21)$$

In Eq. (9.21) we have defined

$$R = Q_2 + \Gamma^T S(k+1)\Gamma$$

for convenience. If we now substitute Eq. (9.20) into Eq. (9.15) for $\lambda(k)$ and $\lambda(k+1)$, we eliminate λ . Then we substitute Eq. (9.21) into Eq. (9.16) to eliminate $x(k+1)$ as follows. From Eq. (9.15), we have

$$\lambda(k) = \Phi^T \lambda(k+1) + Q_1 x(k),$$

and substituting Eq. (9.20), we have

$$S(k)x(k) = \Phi^T S(k+1)x(k+1) + Q_1 x(k).$$

Now we use Eq. (9.16) for $\mathbf{x}(k+1)$

$$\mathbf{S}(k)\mathbf{x}(k) = \Phi^T \mathbf{S}(k+1)(\Phi \mathbf{x}(k) + \Gamma \mathbf{u}(k)) + \mathbf{Q}_1 \mathbf{x}(k).$$

Next we use Eq. (9.21) for $\mathbf{u}(k)$ in the above

$$\mathbf{S}(k)\mathbf{x}(k) = \Phi^T \mathbf{S}(k+1)[\Phi \mathbf{x}(k) - \Gamma \mathbf{R}^{-1} \Gamma^T \mathbf{S}(k+1) \Phi \mathbf{x}(k)] + \mathbf{Q}_1 \mathbf{x}(k),$$

and collect all terms on one side

$$[\mathbf{S}(k) - \Phi^T \mathbf{S}(k+1) \Phi + \Phi^T \mathbf{S}(k+1) \Gamma \mathbf{R}^{-1} \Gamma^T \mathbf{S}(k+1) \Phi - \mathbf{Q}_1] \mathbf{x}(k) = 0. \quad (9.22)$$

Because Eq. (9.22) must hold for any $\mathbf{x}(k)$, the coefficient matrix must be identically zero, from which follows a backward difference equation describing the solution of $\mathbf{S}(k)$

Riccati equation

$$\mathbf{S}(k) = \Phi^T [\mathbf{S}(k+1) - \mathbf{S}(k+1) \Gamma \mathbf{R}^{-1} \Gamma^T \mathbf{S}(k+1)] \Phi + \mathbf{Q}_1, \quad (9.23)$$

which is often rewritten as

$$\mathbf{S}(k) = \Phi^T \mathbf{M}(k+1) \Phi + \mathbf{Q}_1, \quad (9.24)$$

where

$$\mathbf{M}(k+1) = \mathbf{S}(k+1) - \mathbf{S}(k+1) \Gamma [\mathbf{Q}_2 + \Gamma^T \mathbf{S}(k+1) \Gamma]^{-1} \Gamma^T \mathbf{S}(k+1). \quad (9.25)$$

Equation (9.23) is called the discrete **Riccati equation**. It is not easy to solve because it is nonlinear in $\mathbf{S}$. But note that the matrix to be inverted in Eq. (9.25) $\mathbf{R}$, has the same dimension as the number of controls, which is usually less than the number of states.

The boundary condition on the recursion relationship for $\mathbf{S}(k+1)$ is obtained from Eq. (9.19) and Eq. (9.20); thus

$$\mathbf{S}(N) = \mathbf{Q}_1, \quad (9.26)$$

and we see now that the problem has been transformed so that the solution is described by the recursion relations Eq. (9.24) and Eq. (9.25) with the single boundary condition given by Eq. (9.26). The recursion equations must be solved backwards because the boundary condition is given at the endpoint. To solve for $\mathbf{u}(k)$, we use Eq. (9.21) to obtain

$$\mathbf{u}(k) = -\mathbf{K}(k)\mathbf{x}(k), \quad (9.27)$$

where

$$\mathbf{K}(k) = [\mathbf{Q}_2 + \Gamma^T \mathbf{S}(k+1) \Gamma]^{-1} \Gamma^T \mathbf{S}(k+1) \Phi \quad (9.28)$$

and is the desired "optimal" time-varying feedback gain.

Let us now summarize the entire procedure:

1. Let $\mathbf{S}(N) = \mathbf{Q}_1$ and $\mathbf{K}(N) = \mathbf{0}$.
2. Let $k = N$.

3. Let $\mathbf{M}(k) = \mathbf{S}(k) - \mathbf{S}(k)\mathbf{\Gamma}[\mathbf{Q}_2 + \mathbf{\Gamma}^T\mathbf{S}(k)\mathbf{\Gamma}]^{-1}\mathbf{\Gamma}^T\mathbf{S}(k)$.
4. Let $\mathbf{K}(k-1) = [\mathbf{Q}_2 + \mathbf{\Gamma}^T\mathbf{S}(k)\mathbf{\Gamma}]^{-1}\mathbf{\Gamma}^T\mathbf{S}(k)\mathbf{\Phi}$.
5. Store $\mathbf{K}(k-1)$.
6. Let $\mathbf{S}(k-1) = \mathbf{\Phi}^T\mathbf{M}(k)\mathbf{\Phi} + \mathbf{Q}_1$.
7. Let $k = k-1$.
8. Go to step 3.

For any given initial condition for $\mathbf{x}$, to apply the control, we use the stored gains $\mathbf{K}(k)$ and

$$\mathbf{x}(k+1) = \mathbf{\Phi}\mathbf{x}(k) + \mathbf{\Gamma}\mathbf{u}(k), \quad [9.16]$$

where

$$\mathbf{u}(k) = -\mathbf{K}(k)\mathbf{x}(k). \quad [9.27]$$

Note that the optimal gain, $\mathbf{K}(k)$, changes at each time step but can be pre-computed and stored for later use as long as the length, N , of the problem is known. This is so because no knowledge of the initial state $\mathbf{x}(0)$ is required for computation of the control gain $\mathbf{K}(k)$.

◆ Example 9.3 Time-Varying Nature of Control Gains

Solve for the time history of $\mathbf{K}$ for the satellite attitude-control example described in Appendix A.1. Choose the state weighting matrix to be

$$\mathbf{Q}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad (9.29)$$

which means that the angle state is weighted but not the angular velocity. Choose the control weighting matrix, a scalar in this case because there is a single control input, to have three values

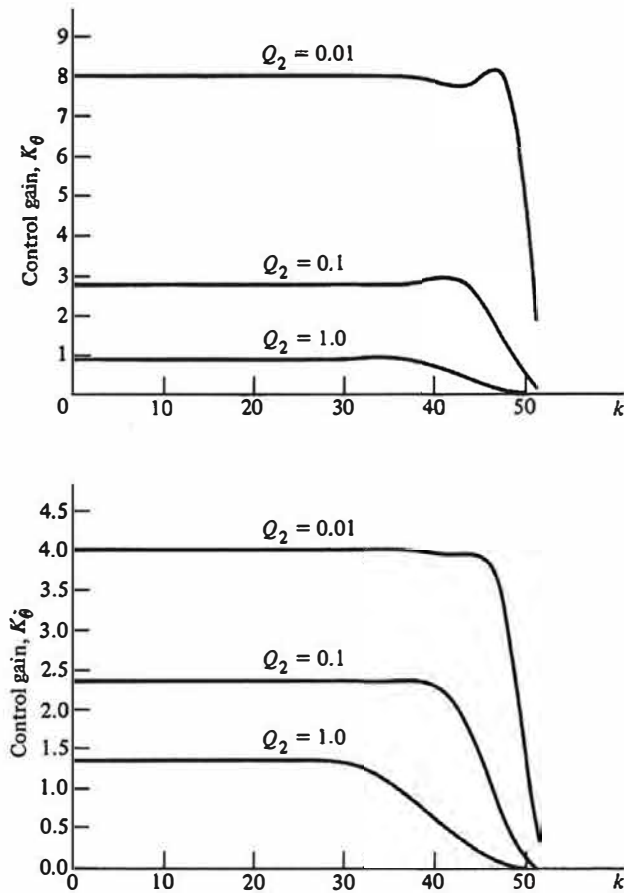
$$\mathbf{Q}_2 = 0.01, \quad 0.1, \quad \text{and} \quad 1.0, \quad (9.30)$$

and plot the resulting time histories of $\mathbf{K}$.

Solution. Equations (9.24) through (9.28) need to be solved for the system transfer function $G(s) = 1/s^2$. The problem length for purposes of defining $\mathcal{J}$ was chosen to be 51 steps, which, with the sample period of $T = 0.1$ sec. means that the total time was 5.1 sec. This time was chosen long enough so that it was apparent that the gains were essentially constant over the initial time period.

Figure 9.3 contains the resulting gain time histories plotted by the computer. We see from the figure that the problem length affects only the values of K near the end, and in fact, the first portions of all cases show constant values of the gains. If the problem length had been

Figure 9.3
Example of control gains
versus time, Example 9.3



chosen to be longer, the end characteristics would have been identical, and the early constant gain portion would have existed for a longer time.

The fact that the gain over the first portion of the example was constant is typical of the optimal gains for all constant coefficient systems, provided that the problem time is long enough. This means that the optimal controller over the early, constant-gain portion is identical to the constant-gain cases discussed in Chapter 8 and Section 9.1 except that the values of the constant gain, $\mathbf{K}$, are based on the minimization of a cost function rather than a computation based on specified root locations. We could also view this result as a method to find

a “good” set of pole locations because the constant, optimal $\mathbf{K}$ during the early portion of the solution determines a set of closed-loop roots.

For MIMO problems, the time-varying gains act exactly as in the preceding example. The next section develops a method to compute the constant value of the optimal gains so that they can be used in place of the time-varying values, thus yielding a much simpler implementation. The only region where the gains are not optimal is during the transient region near the end. In fact, many control systems are turned on and left to run for very long times; for example, a satellite-attitude control system might run for years. This kind of problem is treated mathematically as if it runs for an infinite time, and therefore the constant-gain portion of the time-varying optimal solution *is* the true optimal.

Before we leave the time-varying case it is informative to evaluate the optimal cost function $\mathcal{J}$ in terms of λ and $\mathbf{S}$. If we substitute Eqs. (9.13) and (9.14) for $\lambda^T(k+1)\Gamma$ and $\lambda^T(k+1)\Phi$ in Eq. (9.12), we find

$$\begin{aligned}\mathcal{J}' &= \frac{1}{2} \sum_{k=0}^N [\mathbf{x}^T(k)\mathbf{Q}_1\mathbf{x}(k) + \mathbf{u}^T(k)\mathbf{Q}_2\mathbf{u}(k) - \lambda^T(k+1)\mathbf{x}(k+1) \\ &\quad + (\lambda^T(k) - \mathbf{x}^T(k)\mathbf{Q}_1)\mathbf{x}(k) + (-\mathbf{u}^T(k)\mathbf{Q}_2)\mathbf{u}(k)] \\ &= \frac{1}{2} \sum_{k=0}^N [\lambda^T(k)\mathbf{x}(k) - \lambda^T(k+1)\mathbf{x}(k+1)] \\ &= \frac{1}{2} \lambda^T(0)\mathbf{x}(0) - \frac{1}{2} \lambda^T(N+1)\mathbf{x}(N+1).\end{aligned}$$

However, from Eq. (9.19), $\lambda(N+1) = \mathbf{0}$, and thus, using Eq. (9.20), we find

$$\begin{aligned}\mathcal{J}' = \mathcal{J} &= \frac{1}{2} \lambda^T(0)\mathbf{x}(0) \\ &= \frac{1}{2} \mathbf{x}^T(0)\mathbf{S}(0)\mathbf{x}(0).\end{aligned}\tag{9.31}$$

Thus we see that having computed $\mathbf{S}$, we can immediately evaluate the cost associated with the control. Although the cost could be used in evaluating different candidate designs, in fact, it is not very useful because the weighting matrices, $\mathbf{Q}_1$ and $\mathbf{Q}_2$, are arbitrary quantities that change with the different designs. Furthermore, the value of the discrete cost as defined by Eq. (9.11) is roughly proportional to the sample rate, thus eliminating any usefulness in evaluating the performance for different sample rates. Typically a designer would evaluate different designs by looking at the traditional measures that have been discussed in Chapter 7 or possibly by evaluating quadratic performance measures that are independent of the $\mathbf{Q}$'s and the sample rate. We will show in Section 9.3.4 how to remove the effect of sample rate on the value of the cost.

9.3 LQR Steady-State Optimal Control

regulator

LQR

The previous section developed the optimal control gain that minimized the cost in Eq. (9.11). We saw that the result was a time-varying gain, $\mathbf{K}(k)$, but that there would usually be a portion of the solution that produced a constant gain, $\mathbf{K}_\infty$, which would be much easier to implement in a control system. In fact, for the infinite time problem, called the **regulator** case, the constant-gain solution is the optimum. We call this solution the **linear quadratic regulator**, or LQR, because it applies to *linear* systems, the cost is *quadratic*, and it applies to the *regulator* case. This section will discuss how one finds the LQR solution and various properties of the solution.

One obvious method to compute the value of $\mathbf{K}$ during the early, constant portion of a problem is to compute $\mathbf{S}$ backward in time until it reaches a steady value, $\mathbf{S}_\infty$, then use Eq. (9.28) to compute $\mathbf{K}_\infty$. This has been done in some software packages and gives reliable answers. Its drawback is that it requires substantially more computation than the alternate methods.

Another method is to look for steady-state solutions of the Riccati equation. In steady state, $\mathbf{S}(k)$ becomes equal to $\mathbf{S}(k+1)$ (we'll call them both $\mathbf{S}_\infty$) and the Riccati Eq. (9.23) reduces to

$$\mathbf{S}_\infty = \Phi^T [\mathbf{S}_\infty - \mathbf{S}_\infty \Gamma \mathbf{R}^{-1} \Gamma^T \mathbf{S}_\infty] \Phi + \mathbf{Q}_1, \quad (9.32)$$

algebraic Riccati equation

which is usually referred to as the **algebraic Riccati equation**. Because of the quadratic appearance of $\mathbf{S}_\infty$, there is more than one solution, and one needs to know that $\mathbf{S}$ must be positive definite to select the correct one. The fact that $\mathbf{S}$ is positive definite follows by inspection of Eq. (9.31) and that $\mathcal{J}$ must be positive. For extremely simple problems, one is sometimes able to use Eq. (9.32) to find an analytical solution for $\mathbf{S}_\infty$, but in most cases this is impossible, and a numerical solution is required.

Euler-Lagrange equations

Most software packages use variations on a method called **eigenvector decomposition** due to its superior computational efficiency compared to the methods above. It is based on the linear description of the combined state and adjoint equations given by Eqs. (9.16), (9.17), and (9.18), which describe the time-varying solution. We can combine these equations into a set of difference equations in standard form in $\mathbf{x}$ and $\boldsymbol{\lambda}$ if we assume that $\mathbf{Q}_2$ and Φ are nonsingular.³ These equations are called **Hamilton's equations** or the **Euler-Lagrange equations**

$$\begin{bmatrix} \mathbf{x} \\ \boldsymbol{\lambda} \end{bmatrix}_{k+1} = \begin{bmatrix} \Phi + \Gamma \mathbf{Q}_2^{-1} \Gamma^T \Phi^{-T} \mathbf{Q}_1 & -\Gamma \mathbf{Q}_2^{-1} \Gamma^T \Phi^{-T} \\ -\Phi^{-T} \mathbf{Q}_1 & \Phi^{-T} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\lambda} \end{bmatrix}_k \quad (9.33)$$

³ For systems with a pure time delay that is greater than the sampling period, $\mathbf{T}$, Φ is singular and the following development would fail. Software packages usually have features in their formulations that circumvent this difficulty.

and their system matrix is called the control **Hamiltonian** matrix

$$\mathcal{H}_c = \begin{bmatrix} \Phi + \Gamma Q_2^{-1} \Gamma^T \Phi^{-T} Q_1 & -\Gamma Q_2^{-1} \Gamma^T \Phi^{-T} \\ -\Phi^{-T} Q_1 & \Phi^{-T} \end{bmatrix}. \quad (9.34)$$

Because the system described by Eq. (9.33) is linear and the Hamiltonian matrix is constant, we can solve for the eigenvalues of Eq. (9.34) [or roots of Eq. (9.33)] using standard techniques (see `eig.m` in `MATLAB`). For an n th order system, there will be $2n$ eigenvalues. We will show in the next section that n of the roots are stable and the other n are unstable. In fact, the n unstable roots are the reciprocals of the n stable roots. Furthermore, the n stable roots are the roots of the optimal, constant-gain, closed-loop system! If we were trying to find the optimal $\mathbf{K}$ for a SISO system, the problem would now be complete, because knowing the optimal roots allows us to use Ackermann's formula to find $\mathbf{K}$. But we want the optimal $\mathbf{K}$ for MIMO systems, too, so it's not that simple. We will return to the eigenvector decomposition solution after establishing the characteristics of the roots just stated.

9.3.1 Reciprocal Root Properties

Let us turn to the question of the reciprocal nature of the roots of Eq. (9.33). If we take the z -transforms of Eqs. (9.16), (9.17), and (9.15), we obtain

$$z\mathbf{X}(z) = \Phi\mathbf{X}(z) + \Gamma\mathbf{U}(z), \quad (9.35)$$

$$\mathbf{U}(z) = -z\mathbf{Q}_2^{-1}\Gamma^T\Lambda(z), \quad (9.36)$$

$$\Lambda(z) = \mathbf{Q}_1\mathbf{X}(z) + z\Phi^T\Lambda(z). \quad (9.37)$$

If we substitute Eq. (9.36) into Eq. (9.35) and write the remaining two equations in terms of the variables $\mathbf{X}(z)$ and $z\Lambda(z)$, we find, in matrix form

$$\begin{bmatrix} z\mathbf{I} - \Phi & \Gamma\mathbf{Q}_2^{-1}\Gamma^T \\ -\mathbf{Q}_1 & z^{-1}\mathbf{I} - \Phi^T \end{bmatrix} \begin{bmatrix} \mathbf{X}(z) \\ z\Lambda(z) \end{bmatrix} = [\mathbf{0}].$$

Thus the roots of the Hamiltonian system are those values of z for which

$$\det \begin{bmatrix} z\mathbf{I} - \Phi & \Gamma\mathbf{Q}_2^{-1}\Gamma^T \\ -\mathbf{Q}_1 & z^{-1}\mathbf{I} - \Phi^T \end{bmatrix} = 0.$$

If we now reduce the term $-\mathbf{Q}_1$ to zero by adding $\mathbf{Q}_1(z\mathbf{I} - \Phi)^{-1}$ times the first rows to the second rows, we have

$$\det \begin{bmatrix} z\mathbf{I} - \Phi & \Gamma\mathbf{Q}_2^{-1}\Gamma^T \\ 0 & z^{-1}\mathbf{I} - \Phi^T + \mathbf{Q}_1(z\mathbf{I} - \Phi)^{-1}\Gamma\mathbf{Q}_2^{-1}\Gamma^T \end{bmatrix} = 0.$$

Because this matrix is blockwise triangular, we have

$$\det(z\mathbf{I} - \Phi)\det\{z^{-1}\mathbf{I} - \Phi^T + \mathbf{Q}_1(z\mathbf{I} - \Phi)^{-1}\Gamma\mathbf{Q}_2^{-1}\Gamma^T\} = 0.$$

Now we factor the term $z^{-1}\mathbf{I} - \Phi^T$ from the second term to find

$$\det(z\mathbf{I} - \Phi)\det\{(z^{-1}\mathbf{I} - \Phi^T)\{\mathbf{I} + (z^{-1}\mathbf{I} - \Phi^T)^{-1}\mathbf{Q}_1(z\mathbf{I} - \Phi)^{-1}\Gamma\mathbf{Q}_2^{-1}\Gamma^T\}\} = 0.$$

To simplify the notation, we note that $\det(z\mathbf{I} - \Phi) = a(z)$, the plant characteristic polynomial, and $\det(z^{-1}\mathbf{I} - \Phi^T) = a(z^{-1})$. Thus, using the fact that $\det \mathbf{AB} = \det \mathbf{A} \det \mathbf{B}$, we find that the Hamiltonian characteristic equation is

$$a(z)a(z^{-1})\det\{\mathbf{I} + \rho(z^{-1}\mathbf{I} - \Phi^T)^{-1}\mathbf{H}^T\mathbf{H}(z\mathbf{I} - \Phi)^{-1}\overline{\Gamma\Gamma}^T\} = 0. \quad (9.38)$$

where $\mathbf{Q}_1 = \rho\mathbf{H}^T\mathbf{H}$ and $\Gamma\mathbf{Q}_2^{-1}\Gamma^T = \overline{\Gamma\Gamma}^T$. Now we use the result (Eq. C.3) from Appendix C for the determinant of a sum of $\mathbf{I}$ and a matrix product $\mathbf{AB}$, choosing $\mathbf{A} = (z^{-1}\mathbf{I} - \Phi^T)^{-1}\mathbf{H}^T$ to write

$$a(z)a(z^{-1})\det[1 + \rho\mathbf{H}(z\mathbf{I} - \Phi)^{-1}\overline{\Gamma\Gamma}^T(z^{-1}\mathbf{I} - \Phi^T)^{-1}\mathbf{H}^T] = 0. \quad (9.39)$$

If we replace z by z^{-1} in Eq. (9.39), the result is unchanged because $\det \mathbf{A}^T = \det \mathbf{A}$. Therefore, if z_i is a characteristic root of the optimal system, so is the reciprocal z_i^{-1} , and the desired relationship has been established.

These $2n$ roots are those of the coupled $\mathbf{x}, \lambda$ system described by Eq. (9.33), which describes the solution of the time-varying gain case. But the time-varying gain solution includes the portion where the gains are constant. Furthermore, during the constant-gain portion, the system can be described by Eq. (9.16) with

$$\mathbf{u}(k) = -\mathbf{K}_\infty \mathbf{x}(k),$$

and the roots of this simplified n th order description must be n of the $2n$ roots of Eq. (9.33). But which ones are they? The answer must be the n stable ones, because if any unstable roots were included, the value of $\mathcal{J}$ would be approaching infinity and would be far from optimal.

Therefore we see that once the roots of Eq. (9.33) are found, the n stable ones are the roots of the optimal *constant-gain* case.

9.3.2 Symmetric Root Locus

An interesting special case of Eq. (9.39) occurs for SISO systems. In this case, the cost function $\mathcal{J}$ can be written as $\mathcal{J} = \rho y^2 + u^2$ where $y = \mathbf{H}\mathbf{x}$ and $\mathbf{Q}_2$ was set equal to 1. Therefore, $\overline{\Gamma} = \Gamma$ and we see that $\mathbf{H}(z\mathbf{I} - \Phi)^{-1}\Gamma$ is the plant transfer function $G(z)$. Eq. (9.39) reduces to

$$1 + \rho G(z^{-1})G(z) = 0, \quad (9.40)$$

and is the characteristic equation for SISO optimal control. It is an equation in root-locus form with respect to ρ , the parameter that reflects the relative weighting on output error y and control u . If ρ is small, the optimal roots are near the poles of the plant (or the stable reflections of the poles if $G(z)$ is unstable), and as ρ gets large, the roots go toward the zeros of $G(z^{-1})G(z)$, which are inside the unit circle.

◆ **Example 9.4** *Symmetric Root Locus for Satellite Attitude Control*

Draw the symmetric root locus for the satellite attitude-control problem in Example 4.11 for $T = 1.4$ sec. Comment on the damping of the optimal controller vs. ρ .

Solution. The discrete transfer function from Example 4.11 is

$$G(z) = \frac{(z+1)}{(z-1)^2}.$$

Replacing all the z 's with z^{-1} and multiplying top and bottom by z^2 results in

$$G(z^{-1}) = \frac{z(z+1)}{(z-1)^2}.$$

Therefore, the locus of optimal root locations versus the parameter ρ is determined by substituting the two G 's into Eq. (9.40) to arrive at

$$1 + \rho \frac{z(z+1)^2}{(z-1)^4} = 0.$$

The locus is drawn in Fig. 9.4. Note how the stable locus segments start from the open-loop poles and terminate at the zeros. Also note that, for each stable root, there is an unstable one that is its reciprocal.

The optimal damping ratio for any ρ is $\zeta \cong 0.7$. Designers have always known that picking $\zeta \cong 0.7$ produced a good compromise between speed of response, overshoot, and use of control; it also turns out, for this example, to be the *optimal* solution. This result makes sense because the optimal formulation is attempting to do the same thing that designers have always tried to do, that is, find the roots that achieve a good balance between the output error and the use of control.

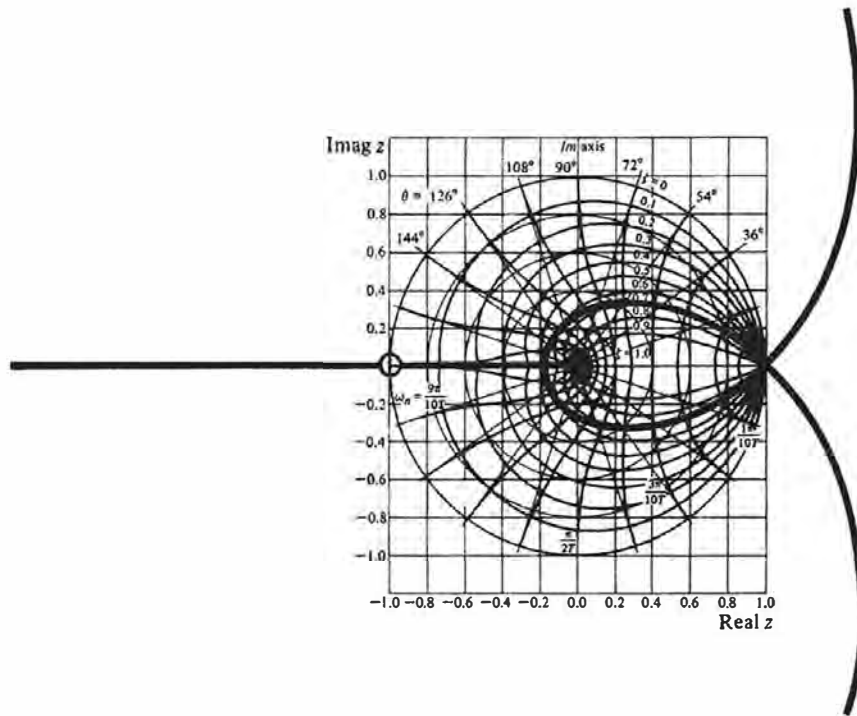
9.3.3 Eigenvector Decomposition

Now let us return to the optimal constant-gain solution for the general case. We can solve Eq. (9.33) by transforming to a new state that has a diagonal system matrix, and from this solution we can obtain the steady-state optimal control. Just as before, the eigenvalues of this matrix are such that the reciprocal of every eigenvalue is also an eigenvalue. Therefore, half the roots of the characteristic equation must be inside the unit circle and half must be outside. In this case, therefore, $\mathcal{H}_c$ can be diagonalized to the form⁴

$$\mathcal{H}_c^* = \begin{bmatrix} \mathbf{E}^{-1} & 0 \\ 0 & \mathbf{E} \end{bmatrix},$$

⁴ In rare cases, $\mathcal{H}_c$ will have repeated roots and cannot be made diagonal by a change of variables. In those cases, a small change in $\mathbf{Q}_1$ or $\mathbf{Q}_2$ will remove the problem, or else we must compute the Jordan form for $\mathcal{H}_c^*$ [see Strang (1976)].

Figure 9.4
Symmetric root locus of Example 9.4



where $\mathbf{E}$ is a diagonal matrix of the unstable roots ($|z| > 1$) and $\mathbf{E}^{-1}$ is a diagonal matrix of the stable roots ($|z| < 1$). $\mathcal{H}_c^*$ is obtained by the similarity transformation

$$\mathcal{H}_c^* = \mathbf{W}^{-1} \mathcal{H}_c \mathbf{W},$$

where $\mathbf{W}$ is the matrix of eigenvectors of $\mathcal{H}_c$ and can be written in block form as

$$\mathbf{W} = \begin{bmatrix} \mathbf{X}_1 & \mathbf{X}_0 \\ \mathbf{\Lambda}_1 & \mathbf{\Lambda}_0 \end{bmatrix},$$

where

$$\begin{bmatrix} \mathbf{X}_0 \\ \mathbf{\Lambda}_0 \end{bmatrix}$$

is the matrix of eigenvectors associated with the eigenvalues (roots) outside the unit circle and

$$\begin{bmatrix} \mathbf{X}_I \\ \mathbf{\Lambda}_I \end{bmatrix}$$

is the matrix of eigenvectors associated with the eigenvalues of $\mathcal{H}_c$ that are inside the unit circle.

This same transformation matrix, $\mathbf{W}$, can be used to transform $\mathbf{x}$ and $\boldsymbol{\lambda}$ to the normal modes of the system, that is,

$$\begin{bmatrix} \mathbf{x}^* \\ \boldsymbol{\lambda}^* \end{bmatrix} = \mathbf{W}^{-1} \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\lambda} \end{bmatrix},$$

where $\mathbf{x}^*$ and $\boldsymbol{\lambda}^*$ are the normal modes. Conversely, we also have

$$\begin{bmatrix} \mathbf{x} \\ \boldsymbol{\lambda} \end{bmatrix} = \mathbf{W} \begin{bmatrix} \mathbf{x}^* \\ \boldsymbol{\lambda}^* \end{bmatrix} = \begin{bmatrix} \mathbf{X}_I & \mathbf{X}_0 \\ \mathbf{\Lambda}_I & \mathbf{\Lambda}_0 \end{bmatrix} \begin{bmatrix} \mathbf{x}^* \\ \boldsymbol{\lambda}^* \end{bmatrix}. \quad (9.41)$$

The solution to the coupled set of difference equations Eq. (9.33) can be simply stated in terms of the initial and final conditions and the normal modes, because the solution for the normal modes is given by

$$\begin{bmatrix} \mathbf{x}^* \\ \boldsymbol{\lambda}^* \end{bmatrix}_N = \begin{bmatrix} \mathbf{E}^{-N} & \mathbf{0} \\ \mathbf{0} & \mathbf{E}^N \end{bmatrix} \begin{bmatrix} \mathbf{x}^* \\ \boldsymbol{\lambda}^* \end{bmatrix}_0. \quad (9.42)$$

To obtain the steady state, we let N go to infinity; therefore $\mathbf{x}^*(N)$ goes to zero and, in general, $\boldsymbol{\lambda}^*(N)$ would go to infinity because each element of $\mathbf{E}$ is greater than one. So we see that the only sensible solution for the steady-state ($N \rightarrow \infty$) case is for $\boldsymbol{\lambda}^*(0) = \mathbf{0}$ and therefore $\boldsymbol{\lambda}^*(k) = \mathbf{0}$ for all k .⁵

From Eqs. (9.41) and (9.42) with $\boldsymbol{\lambda}^*(k) \equiv \mathbf{0}$, we have

$$\mathbf{x}(k) = \mathbf{X}_I \mathbf{x}^*(k) = \mathbf{X}_I \mathbf{E}^{-k} \mathbf{x}^*(0), \quad (9.43)$$

$$\boldsymbol{\lambda}(k) = \mathbf{\Lambda}_I \mathbf{x}^*(k) = \mathbf{\Lambda}_I \mathbf{E}^{-k} \mathbf{x}^*(0). \quad (9.44)$$

Therefore Eq. (9.43) leads to

$$\mathbf{x}^*(0) = \mathbf{E}^k \mathbf{X}_I^{-1} \mathbf{x}(k). \quad (9.45)$$

Thus, from Eqs. (9.44) and (9.45)

$$\boldsymbol{\lambda}(k) = \mathbf{\Lambda}_I \mathbf{X}_I^{-1} \mathbf{x}(k) = \mathbf{S}_\infty \mathbf{x}(k),$$

which is the same form as our assumption Eq. (9.20) for the sweep method, so we conclude that

$$\mathbf{S}_\infty = \mathbf{\Lambda}_I \mathbf{X}_I^{-1} \quad (9.46)$$

⁵ From Eq. (9.41) we see that if $\boldsymbol{\lambda}^*$ is not zero then the state $\mathbf{x}$ will grow in time and the system will be unstable. However, if the system is controllable we know that a control exists which will make the system stable and give a finite value to $\mathcal{J}$. Because we have the optimal control in Eq. (9.33), it must follow that the optimal system is stable and $\boldsymbol{\lambda}^* \equiv \mathbf{0}$ if $\mathbf{Q}_1$ is such that all the states affect $\mathcal{J}$.